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Joint state of health and remaining useful life assessment of lithium-ion batteries using a novel attention mechanism-based deep learning model and multi-source optimized health indicators

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ABSTRACT

Accurate assessment of the State of-Health (SOH) and Remaining Useful Life (RUL) of lithium-ion batteries is a core challenge for ensuring the safe operation of energy storage systems. Recent studies have emphasized the importance of computational efficiency while acknowledging that traditional models still require significant performance improvements. However, many existing joint SOH-RUL methods still rely on a single traditional model or stack multiple conventional models. Therefore, this paper proposes a novel deep learning framework based on a Deep Spatiotemporal Conv-Attention mechanism. It introduces a feature matrix similarity calculation and a lightweight convolution-based feature diversity module, enabling the model to capture both long- and short-term patterns, improve generalization, and reduce computational resource requirements. Furthermore, to preserve key degradation information from multi-sensor data, differential and integral methods are applied to extract various health indicators (HIs), and then an optimized HIs selection algorithm based on Pearson and Spearman correlation coefficients is proposed to screen these indicators. Finally, to comprehensively evaluate the effectiveness, generalization capability, robustness, computational efficiency, and complexity of the proposed method, comparative experiments were conducted against various mainstream state-of-the-art models using three battery datasets with different materials, charging-discharging protocols, and operating temperatures. The experimental results demonstrate that the proposed model achieves superior performance across all battery datasets, with a mean absolute percentage error for SOH below 1.156% and a relative error for RUL within 2.210%, providing an efficient and reliable solution for the joint estimation of SOH and RUL in lithium-ion batteries.

1. Introduction

Lithium-ion batteries are regarded as one of the primary battery types for future applications due to their lightweight, high energy density, and excellent cycling performance, making them highly promising for aviation industry, electric vehicle and other domains [1,2]. Therefore, precise estimation of the lithium-ion battery state-of-health (SOH) is essential for guaranteeing the reliability and safety of battery performance [3–5].

Recent approaches to SOH estimation can generally be classified into model-driven and data-driven methods. [6]. Model-driven approaches simulate the internal chemical reaction mechanisms of batteries, enabling the characterization of failure modes in battery parameters with a certain level of interpretability and prediction accuracy [7,8]. In contrast, data-driven methods avoid the need for developing detailed electrochemical models or extensively analyzing electrochemical reactions and battery aging mechanisms. Instead, they depend on sensor-collected data, such as current, voltage, temperature, and resistance throughout each cycle, to evaluate the present health condition of the battery through data-driven learning, providing generalization potential.

The predominant data-driven methods currently include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), such as Long Short-Term Memory (LSTM) [9], Gated Recurrent Units (GRU) [10], and Transformers [11]. CNNs are adept in extracting local features using convolutional kernels, and reduce the spatial dimensions

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of features through pooling operations [12]. Qian et al. introduced a technique utilizing a one-dimensional CNN to estimate the state-of-health (SOH) of lithium-ion batteries, where random segments of the charging curve serve as the input [13]. The results showed that the proposed method performs well in estimating local capacity growth. RNNs maintain long-term dependencies and temporal dynamic features in time series data by recurrently connecting information from previous time steps. Transformers, employing multi-head self-attention mechanisms to process sequential data, can handle sequences more efficiently in parallel compared to traditional RNNs. Chen et al. applied the Transformer network to SOH estimation for lithium-ion batteries and illustrated that Transformer models outperform RNN-based models, in terms of both prediction performance and computational efficiency [14]. Jia et al. introduced a network that integrates Bidirectional GRU (Bi-GRU) and Transformers, demonstrating its superior accuracy, enhanced robustness, and improved generalization ability compared to the traditional Transformer model [15].

In addition to model performance, the extraction of health indicators (HIs) is also crucial for the accuracy of SOH estimation. The extraction of these indicators primarily depends on various curves during the battery cycling process, such as voltage, current, temperature, and internal resistance variation curves [16]. Although Wang et al. [17] extracted HIs related to the battery's internal resistance, this process typically requires impedance spectrum measurements, which are time-consuming and require specialized instruments. On the other hand, Li et al. [18] extracted multiple features from batteries with different materials under various operating temperatures and charge-discharge rates. However, temperature related HIs are not consistently stable across different operating conditions. In contrast, the IC curve has shown better performance due to its strong correlation with the polarization reactions inside the battery, although data preprocessing is required to obtain high-quality indicators. For example, Li et al. [19] compared the effectiveness of different filtering methods and extracted HIs from various points on the IC curve.

In contrast, time-based HIs extraction is simpler as it does not require differentiation processing. For instance, Wang et al. [1] extracted several statistical features, such as differences and slope standard deviation, from partial segments of charging voltage and current, and combined them with Physics-Informed Neural Networks (PINNs). This method achieved good results across various datasets. Furthermore, HIs based-on time differences are the simplest to extract. For example, Li et al. [20] used features related to constant current charging time as model inputs. This approach only requires the time difference of selected segments and does not require noise reduction of the original data. The authors used the Pearson Correlation Coefficient (PCC) to verify that voltage segments within a small range still have high representational ability. However, the method of extracting HIs from smaller segments is neither comprehensive nor efficient. To address this, Li et al. [2] employed smaller and higher-frequency charging time segments, and used PCC to compress the charging segments, aiming to extract more relevant HIs from shorter charging segments.

While Transformers possess powerful global modeling capabilities, they have insufficient sensitivity to local features, whereas CNNs excel at extracting local features but involve extensive computational operations. Depthwise separable convolutions have been proven to be an effective alternative to traditional CNNs, reducing computational resource consumption through pointwise and depthwise convolutions, thereby enabling rapid estimation of battery SOH.

In the field of lithium-ion battery health management, although numerous studies have focused on predicting battery state-of-health (SOH) and remaining useful life (RUL), most methods remain limited to single-task predictions and lack joint estimation of SOH and RUL. For example, E. Chemali et al. leveraged the strong local feature extraction capability of convolutional neural networks (CNNs) to perform single-task SOH estimation [21], while Z. Li employed a combination of Bi-LSTM and Transformer networks to improve modeling of long-term sequential data [22]. S. Bockrath used a Temporal Convolutional Network (TCN) to directly estimate SOH from partial charge–discharge segments [23], and T. Wang applied a Transformer encoder for feature extraction within a self-supervised framework, also only for SOH prediction [24].

In terms of RUL prediction, existing methods have similarly focused on single-task approaches. Y. Cai combined Transformers with signal decomposition techniques for RUL prediction [25], U. Saleem proposed the TransRUL model specifically for RUL estimation [26], and S. Ly improved LSTM models to construct RUL conditional probability density functions, enabling richer and more accurate RUL predictions [27]. Y. Liu first used CNNs to predict the maximum discharge capacity for each cycle to construct degradation curves, then applied empirical mode decomposition to extract global degradation trends, and finally employed a GRU–fully connected network to complete the RUL prediction task [28].

Joint SOH–RUL assessment remains a challenging task, and existing studies mainly focus on effectively modeling the intrinsic coupling between SOH and RUL and improving prediction accuracy. The core methodological strategies can be summarized from two aspects. First, in terms of joint modeling, shared parameters of the selected model layers are often adopted to learn general degradation representations, or multiple task-specific loss functions are integrated to enable coordinated training. Second, for accuracy enhancement, prior studies tend to increase the representational capacity of input data or employ advanced deep learning architectures, such as CNN and Transformer, as well as hybrid frameworks that combine multiple representative models to strengthen feature extraction and sequential prediction. These model designs are often further complemented by handcrafted feature engineering or data preprocessing techniques, such as denoising algorithms, to improve prediction accuracy and robustness. For example, B. Chen et al. employed a hybrid architecture combining multi-channel CNN, BiLSTM, BiGRU, and attention mechanisms to enhance prediction accuracy. While this approach improves performance, it may also introduce substantial redundancy in model training [29].

Most models are trained on manually processed and selected features, which generally offer higher quality than raw data but may lose some critical degradation information. To address this limitation, some studies adopt data upscaling and convolutional operations to directly extract features from all raw data. For instance, N. Cai transformed battery cycling data into dual-modality images (curve images and Gramian Angular Field images) to capture degradation characteristics at the image level, and further employed a deep network based on depthwise separable convolutions, using feature fusion and interaction modules to directly output the results of SOH and RUL [30]. J. Chen et al. also reconstructed single-cycle charging data, including voltage and current, into 2D feature maps and applied a 2D convolutional neural network (2D-CNN) to extract spatial features related to degradation. These features were then combined with correlation-guided handcrafted features to form multi-scale hybrid representations. Joint optimization of the two targets was achieved through shared network parameters and task-specific loss functions; however, this approach also imposes higher computational demands during training [31].

Designing more refined feature extraction methods may be effective for SOH estimation, however, for large-scale time-series prediction tasks such as RUL prediction, a more comprehensive understanding of global information is required. To address this, R. Zheng applied Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ICEEMDAN) to decompose the data into a series of intrinsic mode functions (IMFs), reducing noise in the raw data and enabling the learning of long-term degradation patterns from low-frequency modes. Key parameters of the Deep Belief Network (DBN) were further optimized using a Bayesian algorithm, resulting in the Optimized Deep Belief Network with Bayesian Algorithm (ODBN-BA) model for end-to-end joint prediction of SOH and RUL [32]. Furthermore, P. Yang et al. proposed a multi-task learning model called Generalized Broad Learning

System (GBLS) Booster, which combines CNN's strength in short-term feature extraction with the Transformer's capability for long-term sequence modeling. In this framework for joint SOH-RUL assessment, features were engineered and denoised using Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), then passed through a shared network layer to learn general degradation representations, followed by task-specific output layers, thereby achieving both high accuracy and strong generalization performance [33].

To address computational complexity and deployment efficiency in prediction, increasing attention has been given to lightweight model architectures and improved attention mechanisms. For instance, K. Lv adopted single-scale depthwise separable convolutions (DSConv) in the feature extraction module, reducing computational overhead while maintaining performance for RUL prediction [34]; D. Zhou proposed a lightweight temporal convolutional (TCN) feature extractor to enable online joint estimation of SOH and RUL, while significantly reducing parameters and computational cost [35]; S. Jafari employed a lightweight CNN-GRU hybrid structure to extract sequential features from charging curves, demonstrating the feasibility of using sequence networks with fewer parameters for RUL prediction [36].

In summary, with the development of computational resources and algorithms, simple single-input data and linear correlation coefficients are insufficient to allow data-driven approaches to potentially replace model-based methods in the future. Therefore, it is necessary to fully consider incorporating multimodal and multi-source inputs as learning material for the model, and to employ multiple correlation evaluation methods or design appropriate algorithms to assist the model in reducing the difficulty of learning battery SOH degradation patterns from raw data. This, to some extent, promotes the advancement of data-driven approaches.

Meanwhile, with the rapid upgrade of computing devices, it has been increasingly demonstrated in many references that simple traditional machine learning models, such as Random Forest Regression and Extreme Learning Machine, struggle to find suitable functions to fit the relationship between inputs of diverse modalities, degradation patterns, and battery SOH. This necessitates state estimation models with more complex network architectures. Moreover, the trade-off between model complexity and accuracy is also a topic worthy of investigation, as it is closely related to maximizing model performance given the available computational resources, particularly on resource-constrained edge devices. Consequently, experiments on model complexity should also be regarded as an evaluation metric for model performance.

In conclusion, the two main challenges in existing methods can be summarized as below:

(1) The raw data collected from multiple sensors contain diverse degradation-related features; however, many existing methods still rely on a single or limited data source. In addition, feature selection is often primarily guided by Pearson (PCC) and/or Spearman (SCC) correlation analyses, which may overlook other informative health indicators and, to some extent, influence the accuracy and generalization of the results.

(2) In recent years, growing attention has been given to improving the computational efficiency of model design, as studies increasingly suggest that traditional models may still have room for performance enhancement. However, many existing joint estimation approaches for SOH and RUL mainly emphasize improving prediction accuracy through the integration of traditional models or optimization algorithms, while giving relatively limited consideration to computational efficiency.

To tackle these challenges, this work introduces a HI selection algorithm that considers the HI representation capabilities across all batteries in a group, and a Deep Spatiotemporal Conv-Attention (DSCA) mechanism for accurate and efficient SOH estimation. The main contributions are concluded as follows:

(1) A multi-task deep learning framework based on DSConv and DSCA is proposed for the joint assessment of SOH and RUL. By reformulating the traditional attention computation, the proposed attention mechanism reduces the computational complexity from $O(N^2d)$ to $O\left(Nd^2\right)$. Since the sequence length N in time-series applications is much larger than the feature dimension d(e.g., $N \gg d$), this reformulation can effectively reduce the overall computational cost by approximately one order of magnitude or more.

(2) To compensate for the reduced diversity caused by the rank limitation in the attention mechanism while maintaining computational efficiency, this work proposes multi-scale DSConv modules, which enhance the model's ability to capture both long-term degradation patterns and short-term local capacity recovery behaviors compared with traditional convolutions.

(3) To ensure effective representation of health indicators (HIs) in multi-source online SOH estimation, a novel health indicator selection algorithm is developed. This method comprehensively evaluates the correlations between various HIs and SOH to achieve standardized feature selection across different battery datasets. In addition, a series of feature comparison experiments, including multiple and hybrid feature inputs, are conducted to validate the robustness and generalization capability of the selected indicators.

(4) To address short-term capacity regeneration and noise interference, the complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) method is applied to decompose the raw capacity curves. The intrinsic mode function (IMF) with the highest correlation is selected to obtain a smoothed and quasi-monotonic trend input. This data-driven approach requires no prior knowledge and significantly improves the stability and accuracy of RUL prediction.

(5) Verification experiments were conducted on multiple battery datasets with different materials, capacities, and charge–discharge protocols to evaluate the proposed model's accuracy, computational efficiency, and generalization capability. The results demonstrate that the proposed framework outperforms several state-of-the-art baseline models in both prediction precision and efficiency.

The comparison between different existing SOH estimation methods is summarized in Table 1. Specifically, it is examined whether these methods can fully utilize various types of output data generated during battery operation, whether more effective HI selection algorithms can be designed (rather than simply relying on linear contrast (LC), PCC, or SCC, whether more efficient models can be developed instead of merely applying models from other domains, and whether experiments on model complexity have been considered for comparison.

Accurate SOH-RUL assessment relies on extracting reliable HIs from battery operation data. In real-world applications, compared to the discharging process, the charging process demonstrates enhanced stability and controllability [39–41], making it more suitable for identifying consistent and representative HIs.

The overall workflow of the proposed method is illustrated in Fig. 1. In the first stage, full life-cycle aging tests are performed on the battery cells to obtain comprehensive operating data. In the second stage, the collected voltage, current, and temperature signals are smoothed, and the corresponding charge–discharge capacity variations are calculated and further decomposed using the CEEMDAN algorithm. The third stage applies differential and integral analysis to process and interpret the data, from which ten HIs are extracted from four characteristic curves. An optimized feature selection algorithm based on the PCC and SCC is then employed to identify the most representative HIs, and both

Table 1
Comparison between different existing SOH estimation methods.

Reference	Source data considered			HI selection method (author-designed)	Assessment model (whether original)	Efficiency considered
	Current	Voltage	Temperature			
[1]	✓	✓	✗	PCC (✗)	PINN (✓)	✗
[2]	✗	✓	✗	PCC (✓)	BMSFormer (✓)	✓
[12]	✗	✓	✓	PCC + SCC (✗)	BiGRU-Transformer (✗)	✗
[15]	✓	✓	✗	PCC (✗)	Transformer (✗)	✗
[16]	✓	✓	✗	LC (✗)	Gaussian filter (✗)	✗
[17]	✗	✓	✗	LC (✗)	Random Forest Regression (✗)	✗
[37]	✓	✓	✓	PCC (✗)	CNN-Transformer (✗)	✓
[38]	✓	✓	✓	PCC (✗)	CNN (✗)	✗
[23]	✗	✓	✗	PCC (✗)	Bi-LSTM (✗)	✗
[25]	✗	✓	✗	PCC + SCC (✗)	LSSVM-AdaBoost (✗)	✗
[31]	✗	✓	✗	PCC (✗)	LSTM (✗)	✗
[32]	✓	✓	✓	PCC (✗)	LSTM (✗)	✗
[45]	✓	✓	✓	PCC + SCC (✓)	Extreme Learning Machine (✗)	✗
Proposed	✓	✓	✓	PCC + SCC (✓)	DSConv-Attention-based (✓)	✓

individual and fused HIs are subsequently validated through comparative experiments. Furthermore, the selected features are segmented and reformatted into the structure required by the attention mechanism before being fed into the model for training and performance evaluation. In the final stage, the trained model parameters are shared between the SOH single-step prediction task and the RUL multi-step prediction task. These tasks are evaluated from different starting points and compared with state-of-the-art models to provide a comprehensive assessment of the proposed method.

The remainder of this paper is organized as follows: Section 2 provides a detailed explanation of the proposed algorithm and highlights its differences from conventional approaches. Section 3 describes the experimental setup and implementation details. Section 4 presents and discusses the results, including feature evaluation, accuracy and generalization analysis, model complexity comparison, and ablation studies. Finally, Section 5 summarizes the main findings and conclusions.

2. Methods

This section introduces the methods used in this study, including the backbone principles and the proposed improvements. It provides a detailed description of the DSCA and DSConv modules, highlighting their principles and enhancements. Additionally, to mitigate the impact of short-term capacity recovery on SOH and RUL assessment accuracy in lithium-ion batteries, the CEEMDAN method is introduced for effective data denoising.

2.1. The designed deep spatiotemporal convolutional attention module

This section describes the backbone and the proposed DSCA module, which enhances feature extraction by integrating convolutional operations with multi-head attention. It overcomes the limitations of traditional attention mechanisms, efficiently capturing both local and global dependencies for SOH and RUL prediction.

2.1.1. The traditional attention mechanism

In recent years, Transformer-based architectures have become the backbone for a wide range of sequence modeling tasks, owing to their ability to capture long-range dependencies through stacked layers of multi-head attention and feedforward neural networks. A typical Transformer encoder is composed of multiple identical layers, each containing a multi-head self-attention module followed by a position-wise feedforward network. Formally, given an input sequence representation $X \in \mathbb{R}^{n \times d}$, the encoder updates the hidden states through a residual connection and normalization strategy:

$$H^{(l)} = \text{LayerNorm}\left(X^{(l-1)} + \text{MHSA}\left(X^{(l-1)}\right)\right), \tag{1}$$

$$X^{(l)} = \text{LayerNorm}\left(H^{(l)} + \text{FFN}\left(H^{(l)}\right)\right), \tag{2}$$

where $X^{(l-1)}$ denotes the input to the l-th layer, MHSA($\cdot$) represents the multi-head self-attention mechanism, and FFN($\cdot$) denotes a feedforward neural network applied independently to each position. By stacking L such layers, the encoder backbone can effectively refine hierarchical representations while preserving global contextual information.

Building on this backbone structure, the traditional multi-head attention mechanism applies multiple self-attention operations in parallel. For each head, the input is linearly projected into query, key, and value representations, which are used to calculate attention weights based on the similarity between elements in the sequence. The weighted sum of the values produces the output of each head. The outputs of all heads are then concatenated and linearly transformed to form the final representation. Formally, for i-th head and l-th layer with input $X^{(l-1)}$:

$$Q_i = X^{(l-1)} W_i^Q, K = X^{(l-1)} W_i^K, V = X^{(l-1)} W_i^V \tag{3}$$

$$A = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) \tag{4}$$

$$H_i = A_i V_i \tag{5}$$

$$\text{MHSA}\left(X^{(l-1)}\right) = \text{Concat}(H_1, H_2, \ldots, H_h) W^O \tag{6}$$

where W_i^Q, W_i^K, W_i^V are the learnable projection matrices, d_k is the key dimension of each head, h is the number of heads, and W^O is the output of projection matrix, and the function $\text{softmax}(z_i) = \exp(z_i)/\sum_j \exp(z_i)$ normalizes the similarity scores into a valid probability distribution by exponentiating each score and dividing it by the sum of all exponentiated values, where i denotes the index of the current element being normalized, j iterates over all elements in the same vector, and z represents the input vector containing all unnormalized scores.

2.1.2. The proposed DSCA mechanism

Traditional multi-head self-attention mechanisms excel at capturing long-range dependencies by computing global correlations across all elements in the sequence. However, this approach relies on Softmax-based attention and sequential matrix multiplications, which often lead to high computational cost and limited efficiency when handling long-term degradation data of batteries. Moreover, conventional multi-head structures may not fully exploit the local spatiotemporal patterns inherent in multi-source sensor signals, potentially constraining the model's ability to capture fine-grained health indicators. To address these limitations, this work introduces the Deep Spatiotemporal Conv-Attention (DSCA) module, which integrates convolutional operations with an optimized attention mechanism. This design enhances the

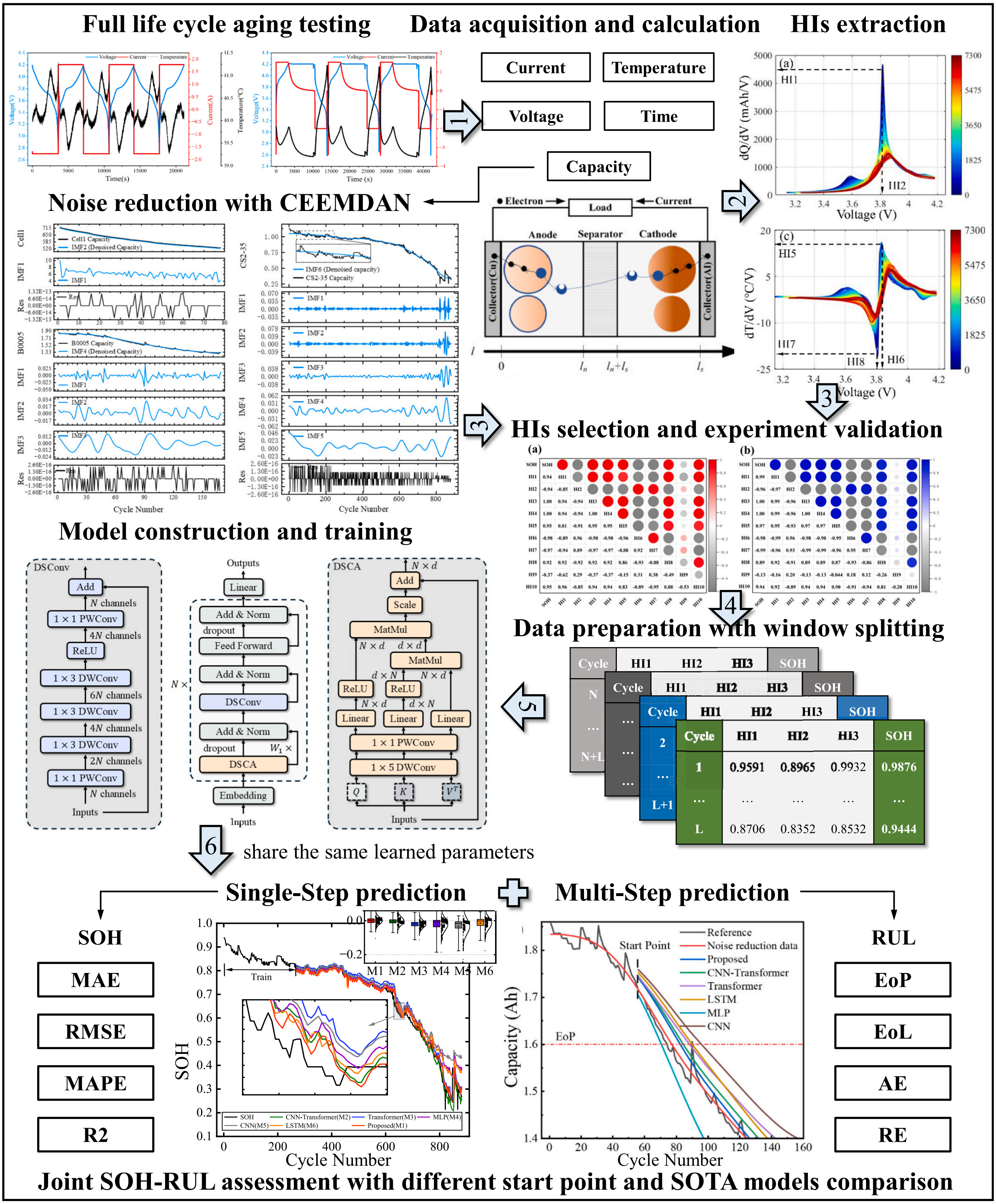

Fig. 1. Flowchart of proposed joint SOH-RUL assessment method.

learning of both local and global dependencies while reducing computational overhead, thereby offering a more efficient and accurate feature extraction framework for SOH and RUL assessment.

The complete model architecture is shown in Fig. 2 and is depicted as follows: initially, the selected HIs (including both multi-source health indicators and the denoised capacity trend extracted via CEEMDAN) serve as input and are processed through the Embedding layer, mapping them into a high-dimensional space for enhanced information extraction. These embedded features are then fed into the DSCA and the Depthwise Separable Convolution (DSConv) modules, which function as shared feature extraction layers to capture both long- and short-term degradation patterns. Addition and layer norm (Add & Norm) layers are incorporated between each module. Finally, the rich representations extracted by these shared layers are passed through two separate Feedforward layers (or output heads), which respectively generate the SOH estimation and RUL prediction as the final outputs of our multi-task model.

For an input x_i to the $i-$th layer in an N-layer model, the corresponding output y_i is computed as follows:

$$x_i' = \mathrm{LayerNorm}(\mathrm{dropout}(\mathrm{DSCA}(x_i)) + x_i) \tag{7}$$

$$x_i' = \mathrm{LayerNorm}(\mathrm{dropout}(\mathrm{DSCA}(x_i)) + x_i) \tag{8}$$

$$x_i'' = \mathrm{LayerNorm}(\mathrm{DSConv}(x_i') + x_i') \tag{9}$$

$$y_i = \mathrm{LayerNorm}(\mathrm{dropout}(\mathrm{FFN}(x_i'')) + x_i'') \tag{10}$$

where DSCA, DSConv, FFN respectively represent the Deep Spatiotemporal Conv-Attention, Depthwise Separable Convolution and Feedforward layer.

To enhance the capacity of the model in capturing both local dependencies and global correlations in sequential data, this study proposes the DSCA module. Unlike conventional self-attention mechanisms that primarily focus on global relationships, DSConv integrates depthwise separable convolutions with a linear attention mechanism, enabling efficient feature extraction while reducing computational complexity [42–44].

The attention mechanism is formulated as follows. First, three linear projections are applied to transform the input into query Q, key K, and value V matrices:

$$Q = \mathrm{ReLU}(XW_Q), K = \mathrm{ReLU}(XW_K), V = XW_V \tag{11}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{E\times E}$ are learnable weight matrices, and the ReLU activation function is applied to Q and K to introduce non-linearity. These matrices are then reshaped to facilitate multi-head attention, allowing for joint modeling across multiple subspaces.

In the proposed DSCA module, a ReLU activation function is applied as the feature mapping operation for the query and key projections. This design is primarily motivated by the requirement of non-negativity in linear attention mechanisms, where the attention weights are computed through kernelized feature mappings rather than the conventional softmax normalization. By enforcing non-negative representations of the projected features, ReLU ensures that the resulting attention scores remain physically interpretable as additive contributions, avoiding undesirable cancellation effects caused by mixed-sign interactions.

From the perspective of numerical stability, the ReLU-based mapping alleviates the sensitivity of attention computation to small-amplitude noise and extreme values, which are commonly encountered in battery degradation data due to measurement uncertainty and short-term capacity fluctuations. By truncating negative responses, high-frequency noise with low signal-to-noise ratios is effectively suppressed, leading to smoother attention responses and more stable optimization behavior during training.

Although the use of ReLU introduces a sparsity constraint that may limit the representational expressiveness of individual attention heads, this effect is mitigated by the use of multiple attention heads, linear projections, and residual connections in the DSCA module. As a result, the overall representational capacity of the model is preserved while benefiting from improved robustness and computational efficiency, which is particularly desirable for lightweight and large-scale sequence modeling tasks.

To incorporate local feature extraction, a depthwise separable convolutional module is employed. The input sequence is first permuted and then processed through a depthwise convolution, which is succeeded by a pointwise convolution.:

$$X_{\mathrm{conv}} = \mathrm{PointwiseConv}\left(\mathrm{DepthwiseConv}\left(X^T\right)\right) \tag{12}$$

where the depthwise convolution extracts fine-grained local dependencies, and the pointwise convolution integrates these features across channels. The resulting representation retains the sequential structure while enriching local information.

Following the convolutional enhancement, the attention computation is performed using a linear attention mechanism. Instead of the traditional SoftMax-based scaling, the interaction between key and value matrices is computed directly:

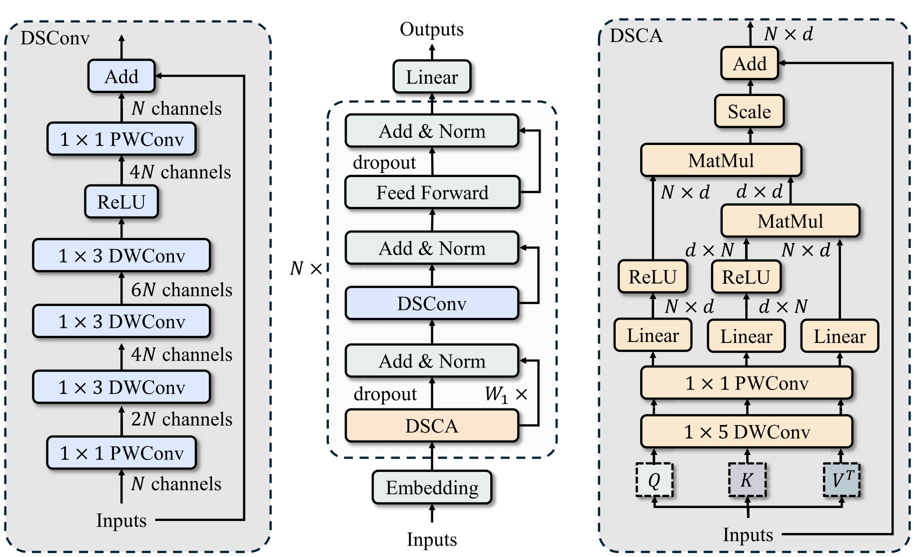

Fig. 2. Proposed model structure.

$$A = KV^T \tag{13}$$

This operation captures intrinsic correlations between different time steps. The attended representation is subsequently obtained by computing the weighted sum of A and the query matrix:

$$Y = \frac{AQ}{\sqrt{d_k}} \tag{14}$$

where d_k represents the dimensionality of each attention head. Finally, a linear transformation is applied, followed by a residual connection that integrates the original input with the attended representation:

$$Z = X + \mathrm{Linear}(Y) \tag{15}$$

This residual connection mitigates gradient vanishing issues and facilitates efficient information propagation. Through the integration of convolutional operations and linear attention, DSCA enhances performance in sequential learning tasks by combining local feature extraction with global context modeling, all while preserving computational efficiency.

To quantitatively evaluate the computational efficiency of the proposed attention mechanism, the complexity of each operation is analyzed and compared with that of the conventional self-attention mechanism.

In the standard multi-head self-attention, the attention matrix $A = QK^T$ requires a pairwise similarity computation between all queries and keys, resulting in a quadratic complexity with respect to the sequence length n:

$$O_{\text{tradition}} = O(n^2 d) \tag{16}$$

where d denotes the embedding dimension. This quadratic dependency severely limits scalability when processing long time-series data.

In contrast, the proposed DSCA reformulates the attention operation into a linearized two-step form. Specifically, it first computes the key–value correlation matrix $M = KV^T \in \mathbb{R}^{d\times d}$, followed by a projection with the query vector:

$$Y = MQ = (KV^T)Q \tag{17}$$

This design eliminates the pairwise token interaction of $O(n^2)$ and reduces the computational complexity to:

$$O_{\text{proposed}} = O(nd^2) \tag{18}$$

Since $d \ll n$ in most practical settings, the proposed structure achieves near-linear complexity with respect to the sequence length. The theoretical ratio between the two complexities is:

$$\frac{O_{\text{proposed}}}{O_{tradition}} == O\left(\frac{d}{n}\right) \tag{19}$$

indicating that DSCA reduces the overall computational cost approximately by a factor n/d.

Moreover, the convolutional enhancement branch introduces an additional term $O(nk)$, where k is the kernel size, which remains negligible compared to $O(nd^2)$.

Therefore, the DSCA mechanism maintains strong spatiotemporal modeling capability while significantly improving computational efficiency and scalability.

2.2. The depthwise separable convolution module

Convolutional operations are a fundamental building block in neural networks, enabling models to extract spatial and temporal dependencies from data. However, standard convolutions often incur high computational cost and memory usage, particularly in time-series applications with long sequences. To balance feature extraction capability and computational efficiency, this section introduces the Depthwise Separable Convolution (DSConv) framework. First, we review conventional and depthwise separable convolutions, highlighting their computational characteristics. Then, we present our specifically designed DSConv module, MBConv1, which leverages multi-layer depthwise convolutions and residual connections to efficiently capture multi-scale local dependencies while maintaining gradient stability.

2.2.1. The conventional convolution and depthwise separable convolution

In convolutional neural networks, a standard convolution layer jointly models spatial correlations and cross-channel interactions through learnable kernels. Specifically, given an input tensor $X \in \mathbb{R}^{B\times H\times W\times C_{in}}$, where B is the batch size, H and W denote the spatial dimensions, and C_{in} is the number of input channels, a convolution kernel $W \in \mathbb{R}^{k\times k\times C_{in}\times C_{out}}$ produces an output feature map Y as

$$Y = X^{*}W \tag{20}$$

where * denotes the 2D convolution operator. For each output channel j, the operation can be expressed asThe computation for each output channel can be expressed as

$$Y_j = \sum_{i=1}^{C_{in}} X_i^{\,*} W_{ij}. \tag{21}$$

The computational complexity of this operation is

$$O(k^2 \cdot C_{in} \cdot C_{out} \cdot H \cdot W), \tag{22}$$

and the number of learnable parameters is

$$P = k^2 \cdot C_{in} \cdot C_{out} \tag{23}$$

Although standard convolutions are effective in jointly capturing local spatial dependencies and inter-channel correlations, they introduce significant computational cost and memory overhead, which becomes a bottleneck in lightweight architectures and time-series applications with long input sequences.

To address this limitation, depthwise separable convolution (DSConv) decomposes the standard convolution into two steps:

First, depthwise convolution applies a $k \times k$ filter to each input channel independently, capturing spatial correlations without mixing channels. Its complexity is

$$O(k^2 \cdot C_{in} \cdot H \cdot W) \tag{24}$$

Next, pointwise convolution employs a 1×1 convolution to linearly combine the outputs across channels to produce C_{out} feature maps, with complexity

$$O(C_{in} \cdot C_{out} \cdot H \cdot W) \tag{25}$$

Thus, the total complexity of DSConv becomes

$$O(k^2 \cdot C_{in} \cdot H \cdot W + C_{in} \cdot C_{out} \cdot H \cdot W), \tag{26}$$

which is significantly lower than the standard convolution when k is large or $C_{\text{out}} \gg C_{\text{in}}$. At the same time, DSConv maintains a comparable ability to capture both spatial and channel dependencies, making it a widely adopted operator in lightweight deep learning models.

2.2.2. The designed DSConv module

A standard convolution is decomposed into a depthwise convolution and a pointwise convolution by DSConv, which breaks down a conventional convolution into two components: a depthwise convolution and a pointwise convolution. [45–48], significantly reducing computational complexity while preserving feature extraction capabilities.

The proposed MBConv1 module is a deep 1D convolutional block designed to capture multi-scale local dependencies while maintaining efficient gradient flow through a residual connection. Given an input tensor $X \in \mathbb{R}^{B\times L\times C}$, where B is the batch size, L is the sequence length, and C is the embedding dimension, the computation begins with a pointwise convolution that expands the channel dimension by a factor of r:

$$Y_1 = X\cdot W^{pw1} \in \mathbb{R}^{B\times L\times rC} \tag{27}$$

where $W^{pw1} \in \mathbb{R}^{C\times rC}$, the expanded feature map is then processed sequentially through three depthwise convolutions with kernel size $k = 3$, which independently operate on each channel to capture local sequential patterns while keeping parameter overhead minimal:

$$Y_{i+1} = \sigma(\mathrm{DepthwiseConv}(Y_i, K)), i = 1, 2, 3 \tag{28}$$

then a ReLU activation is applied to introduce non-linearity:

$$Y_5 = \mathrm{ReLU}(Y_4) \tag{29}$$

after which a second pointwise convolution reduces the channel dimension back to the original size:

$$Y_6 = Y_5\cdot W^{pw2} \in \mathbb{R}^{B\times L\times C} \tag{30}$$

Finally, a residual connection is added to enhance gradient propagation and stabilize training, producing the output

$$Y = X + Y_6 \tag{31}$$

In this manner, MBConv1 efficiently expands the feature space to extract rich representations, leverages depthwise convolutions to capture multi-scale local dependencies, and employs residual connections to maintain training stability, making it well-suited for deep sequential modeling with low computational cost.

2.3. Complete ensemble empirical mode decomposition with adaptive noise

The Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) is an enhanced version of the classical EMD and EEMD methods, specifically designed to address mode mixing and residual noise problems in nonlinear and non-stationary signal analysis. The algorithm incorporates adaptive noise at each stage of the decomposition and averages the results of multiple realizations, ensuring that the extracted intrinsic mode functions (IMFs) exhibit stronger physical interpretability and improved stability.

In this study, CEEMDAN is not treated as a generic preprocessing technique applied to the entire dataset globally, but rather as a task-oriented signal enhancement module with a clear strategic role in RUL prediction that rigorously adheres to the causality principle of time-series analysis. Crucially, the raw capacity sequence is first divided into distinct training and validation sets based on chronological order. CEEMDAN is then applied independently to the capacity data within each of these pre-divided segments (i.e., training data and validation data). This ensures that the feature extraction process for the training data does not incorporate any information from the future (validation) data, thereby strictly preventing data leakage and ensuring practical deploy ability.

Given an observed signal $x(t)$, CEEMDAN first generates perturbed signals by introducing independent realizations of Gaussian white noise, expressed as $x_i(t) = x(t) + \varepsilon\omega_i(t)$, where ε denotes the noise amplitude and $\varepsilon\omega_i(t)$ represents the i-th noise sequence. Each perturbed signal is then decomposed using EMD, and the first IMF obtained from each realization is averaged to yield the final first-level component, i.e.,

$$\mathrm{IMF}_1(t) = \frac{1}{N}\sum_{i=1}^{N}\mathrm{IMF}_{1,i}(t), \tag{32}$$

where N denotes the number of realizations. The corresponding residual is computed as $r_1(t)=x(t) - \mathrm{IMF}_1(t)$.

Subsequent IMFs are extracted iteratively from the updated residuals. At the k-th step, the residual $r_{k-1}(t)$ is perturbed by additive white noise, followed by EMD decomposition. The first IMF of each perturbed signal is averaged to obtain the k-th IMF, written as

$$\mathrm{IMF}_k(t) = \frac{1}{N}\sum_{i=1}^{N}\mathrm{IMF}_{k,i}(t) \tag{33}$$

The residual is updated as $r_k(t) = r_{k-1}(t) - \mathrm{IMF}_k(t)$. The process is repeated until the final residual $r_K(t)$becomes a monotonic function or contains no oscillatory components. The original signal can therefore be reconstructed as

$$x(t) = \sum_{k=1}^{K}\mathrm{IMF}_k(t) + r_K(t) \tag{34}$$

Compared with EMD, CEEMDAN alleviates mode mixing by incorporating noise-assisted decomposition, while simultaneously overcoming the residual ambiguity introduced in EEMD. As a result, each extracted IMF is better defined and more physically meaningful, making CEEMDAN particularly suitable for analyzing nonlinear and non-stationary time series.

In this study, CEEMDAN is not treated as a generic preprocessing technique but is explicitly introduced as a task-oriented signal enhancement module with a clear strategic role in RUL prediction. By decomposing the raw capacity sequence into IMFs with distinct frequency characteristics, CEEMDAN enables the effective isolation of high-frequency noise–dominated components, which are mainly associated with measurement uncertainty and short-term capacity regeneration. These noise-sensitive components can be selectively suppressed during signal reconstruction, thereby reducing irregular fluctuations while preserving the physically meaningful degradation information. Owing to the resulting smoother and quasi-monotonic input representation, the joint SOH–RUL training process exhibits improved convergence behavior, characterized by faster convergence speed, reduced training oscillations, and fewer abrupt prediction jumps, as the denoised signals alleviate gradient instability and facilitate more stable optimization.

Considering the common issues of short-term capacity regeneration and abrupt fluctuations in existing SOH estimation and RUL prediction methods, this study introduces the CEEMDAN technique to denoise and decompose the raw capacity data. The decomposition results of Cell1, B0005 and CS2–35 are presented in Fig. 3.

Based on the previous correlation analysis, HI3, HI4, and HI6 exhibit strong and complementary correlations with SOH, thereby providing a reliable foundation for joint SOH–RUL modeling. However, directly relying on raw capacity curves for RUL prediction is often affected by short-term capacity fluctuations and measurement noise, which reduces prediction accuracy. To address this limitation, CEEMDAN decomposition is employed to extract more representative long-term degradation trends from the capacity sequence.

As shown in Fig. 3, different IMF components capture multi-scale fluctuations of capacity. Among them, higher-order IMFs show the strongest correlation with capacity and can effectively track the overall degradation pattern of SOH. In particular, IMF6 not only suppresses high-frequency disturbances but also preserves the long-term monotonic degradation signal. Therefore, IMF6 is ultimately selected as the “denoised capacity” input feature for the subsequent RUL prediction module, thereby significantly enhancing the accuracy and robustness of lifetime prediction.

2.4. Joint assessment framework

The core of the joint estimation framework is a unified multi-task deep learning model designed to simultaneously predict SOH and RUL. As illustrated in Fig. 2, the model comprises shared feature extraction layers (consisting of DSCA and DSConv modules) that learn

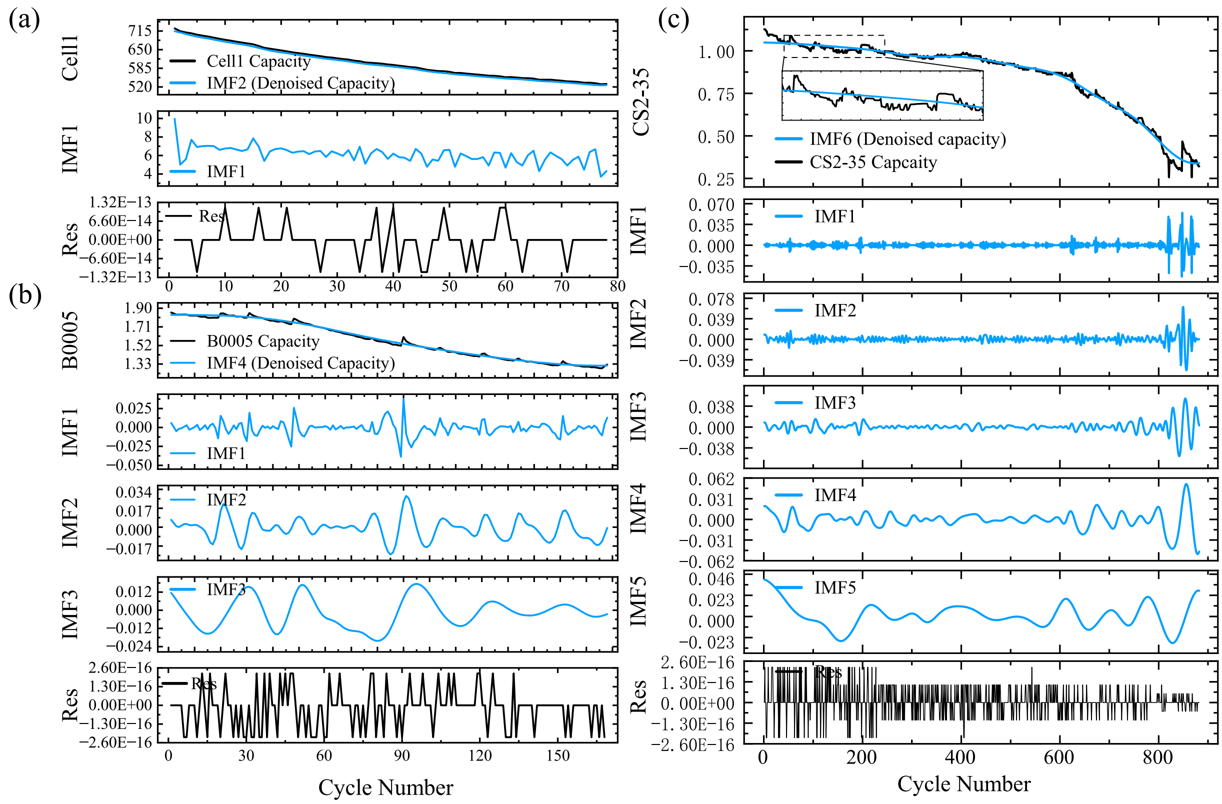

Fig. 3. CEEMDAN decomposition results of (a) Cell1, (b) B0005 and (c) CS2-–35.

comprehensive degradation representations from the input health indicators (including the CEEMDAN-derived denoised capacity). Following these shared layers, the model incorporates two distinct task-specific output heads (Feedforward layers), each responsible for generating the SOH estimation and RUL prediction, respectively. This architecture allows the model to leverage shared knowledge about battery degradation while tailoring predictions to each specific task. The overall optimization objective is to minimize the total loss $\mathscr{L}_{Total}$, defined as:

$$\mathscr{L}_{Total} = \alpha \mathscr{L}_{SOH} + \beta \mathscr{L}_{RUL} \tag{35}$$

In this formulation, $\mathscr{L}_{SOH}$ and $\mathscr{L}_{RUL}$ represent the loss functions for the SOH estimation task and the RUL prediction task, respectively. To ensure training stability and robustness against outliers, both task losses are computed using the Mean Absolute Error (MAE), defined as follows:

$$\mathscr{L}_{SOH} = \mathrm{MAE}(Y_{SOH}, \widehat{Y}_{SOH}) = \frac{1}{n}\sum_{i=1}^{n}\left|Y_{SOH}^{(i)} - \widehat{Y}_{SOH}^{(i)}\right| \tag{36}$$

$$\mathscr{L}_{RUL} = \mathrm{MAE}(Y_{RUL}, \widehat{Y}_{RUL}) = \frac{1}{n}\sum_{i=1}^{n}\left|Y_{RUL}^{(i)} - \widehat{Y}_{RUL}^{(i)}\right| \tag{37}$$

Here, Y denotes the ground truth value, $\widehat{Y}$ represents the predicted value, and n is the number of samples.

The coefficients α and β are crucial weighting parameters that balance the relative importance of the two tasks. Departing from traditional approaches that rely on fixed weights or manual tuning, the innovation of this work is to treat both α and β as trainable parameters. During model training, these weight parameters are updated concurrently with other network parameters (e.g., those in convolutional and LSTM layers) via backpropagation.

This optimization mechanism operates as follows: in the early stages of training, the model dynamically adjusts the values of α and β based on the initial loss magnitudes of the SOH and RUL tasks. If the loss of one task is significantly larger than the other, its corresponding weight may be relatively suppressed. This prevents the gradient of the more challenging task from dominating the overall model update direction in the initial phases, thereby ensuring that both tasks receive adequate learning. Through this dynamic balancing, the model can automatically explore the intrinsic relationship between SOH and RUL predictions and converge to a Pareto-optimal solution—minimizing the total loss $\mathscr{L}_{Total}$ without excessively compromising the performance of either task.

3. Experiments

To ensure a comprehensive and rigorous evaluation of the proposed method, this section provides a detailed description of the experimental setup. First, the evaluation metrics employed in this work are introduced to establish consistent performance criteria. The configurations of the baseline and proposed models are then outlined, followed by the definitions of SOH and RUL to clarify the prediction targets. Subsequently, the datasets used in this study are presented, highlighting their characteristics and relevance to the experiments. Finally, the process of feature extraction and selection is discussed, which serves as the foundation for reliable model training and validation.

3.1. Evaluation metrics

3.1.1. SOH related

Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) are three widely utilized indicators to assess the performance of models [49,50]. The formulas are described as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| \tag{38}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \tag{39}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right| \times 100\% \tag{40}$$

where y represents the actual value, $\widehat{y}$ denotes the predicted value and n is the number of samples.

3.1.2. RUL related

To quantitatively assess the accuracy of RUL prediction, two commonly used error metrics are employed: the Absolute Error (AE) and the Relative Error (RE). AE measures the absolute deviation between the predicted RUL value and the true RUL value. It directly reflects the prediction bias in units of cycles. RE normalizes the absolute error by the true RUL value, thereby providing a scale-independent assessment of prediction accuracy. This is particularly useful when the actual RUL varies significantly across different cells or degradation stages. The formulas for AE and RE are given as follows:

$$\mathrm{AE}_i = |\widehat{RUL}_i - RUL_i| \tag{41}$$

$$\mathrm{RE} = \frac{|\widehat{RUL}_i - RUL_i|}{RUL_i} \tag{42}$$

where $\widehat{RUL}_i$ and RUL_i denote the predicted and true RUL values for the $i-$ th sample, respectively.

3.1.3. Training process

Battery capacity degradation exhibits strong temporal dependency. A single SOH value represents only a static health state at a specific cycle, whereas a short sequence of consecutive cycles implicitly contains dynamic information such as degradation trends and variation rates, including first-order and even second-order temporal characteristics. In the preprocessing stage, a sliding window strategy is adopted to construct sequential inputs, which is widely used in attention-based time-series models. Specifically, a fixed-length window with a length of n and a sliding stride of one cycle is applied to segment the time series composed of HIs and corresponding SOH values.

Due to the sliding window mechanism, the first valid prediction is generated at the n-th cycle rather than at the first cycle. This is not caused by skipping early-cycle information; instead, the model requires a minimum number of historical cycles to form a complete input sequence. Once this initialization condition is satisfied, the SOH is estimated in a one-step-ahead prediction manner, where each prediction corresponds exclusively to the next single cycle.

This sequential formulation allows the model to leverage short-term temporal dependencies while maintaining a consistent single-step SOH estimation setting throughout the prediction horizon.

Prior to model input, the data undergoes min–max normalization to eliminate scale discrepancies among features. For model training, the Adam optimizer is adopted with mean squared error (MSE) as the loss function, formulated as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 \tag{43}$$

Hyperparameter tuning is performed via grid search, and the batch size is empirically set to 128. To mitigate randomness and ensure robustness against data partitioning biases, five independent training and validation runs are conducted under different random seeds in Python. The final test results are averaged across all five test sets to guarantee statistical significance and reproducibility. This rigorous validation framework enhances the reliability of experimental conclusions.

3.2. Model configurations

In this study, a comprehensive comparison of six models is conducted to evaluate the performance of the proposed method. The models considered include the proposed model, the original CNN-Transformer, the conventional Transformer, LSTM, MLP, and a pure CNN model. The proposed model integrates deep separable convolution with attention mechanisms, enabling efficient extraction of local and global features. The CNN-Transformer and Transformer serve as attention-based baselines to investigate the contribution of the proposed dual-channel feature fusion and depthwise separable convolution modules. LSTM is included to represent recurrent architectures commonly used for sequence modeling, while MLP is employed to assess the effect of feed-forward networks without temporal dependencies. The CNN model, on the other hand, provides a representative convolutional baseline to evaluate the capability of purely local feature extraction.

The detailed layer configurations of these models are summarized in Table 2, including embedding dimensions, attention modules, convolutional blocks, normalization layers, pooling, and final prediction layers. This selection of diverse architectures allows for a thorough assessment of both the predictive accuracy and generalization ability of the proposed method relative to conventional deep learning approaches.

Before training, all extracted features from the cycling datasets are normalized using min–max scaling to enhance numerical stability and accelerate convergence. The datasets are then partitioned into training and testing subsets in a 3:7 ratio. To ensure robustness and reproducibility, all reported performance metrics are averaged over five independent experiments with randomized data splits. Subsequently, our proposed model is employed to estimate the state of health from the preprocessed input sequences.

In the training process, a grid search strategy is employed to optimize the hyperparameters of the attention-based models, including the proposed model, CNN-Transformer, and Transformer. Grid search is chosen due to its simplicity and exhaustive nature, which allows systematic evaluation of all possible combinations of predefined hyperparameter values, ensuring that the selected configuration provides robust performance across the validation dataset. The hyperparameters considered in this study include the learning rate (1e-2, 1e-3), number of attention blocks (1, 2, 4), dense layer dimensions (16, 32, 64, 128), embedding dimensions (16, 32, 64, 128), number of attention heads (1, 2, 4), and the number of training epochs (200,1000). By evaluating all combinations of these parameters, the optimal configuration is selected based on the model's performance metrics. In addition, the dense layer dimensions of the MLP model and the hidden layer dimensions of the LSTM are tuned using the same set of dense layer dimensions (16, 32, 64, 128), while the number of layers for both is adjusted via the parameter set (1, 2, 4). For the CNN layers, only the learning rate and the number of training epochs are employed for hyperparameter tuning.

3.3. Multi-task definition

3.3.1. SOH

SOH is a metric used to describe the current condition of a lithium-ion battery relative to its ideal or initial state [39,51–54]. It is commonly characterized as the ratio of the battery's present maximum capacity (C_t) to its initial nominal capacity (C_0), expressed as follows:

$$\mathrm{SOH} = \left(\frac{C_t}{C_0}\right) \times 100\% \tag{44}$$

As the battery undergoes charge and discharge cycles, its capacity gradually declines, leading to a decrease in SOH.

3.3.2. RUL

The task of remaining useful life (RUL) prediction aims to estimate the number of cycles remaining before a battery reaches its End of Performance (EOP). The EOP is defined based on a SOH threshold, typically when $\mathrm{SOH} \leq 0.8$. Let $C(c)$ denote the capacity at cycle c, and define the true EOP index as

Table 2
The layer configurations of different deep learning models.

Proposed	CNN-Transformer	Transformer	MLP	LSTM	CNN
Input	Input	Input	Input	Input	Input
Embed layer	Embed layer	Embed layer	Embed layer	Linear	Conv1D (1 × 5, 1) ReLU
DSCA	Positional Encoder	Positional Encoder	Layer Norm	LSTM layer (32)	MaxPooling
Dropout, Add&Norm	Softmax Attention	Softmax Attention	Dense layer & ReLU	LSTM layer (32)	Conv1D (1 × 7, 1) ReLU
DSConv module	Conv1D (1 × 3, 1) ReLU	Add&Norm	Dense layer& dropout	LSTM layer (32)	MaxPooling
Add&Norm	MaxPooling1D (2,2)	Softmax Attention	Add	Linear	Conv1D (1 × 5, 1) ReLU
MLP	Conv1D (1 × 3, 1) ReLU, Add&Norm	Add&Norm	Linear		Linear
Linear	Softmax Attention	MLP			
	Add&Norm	Add&Norm			
	MLP				
	Add&Norm				
	Linear				

$$EOP = \min\{c \in \mathbb{N} | C(c) \leq \gamma C_0\}. \tag{45}$$

Given an observation cycle c_0, the RUL in cycles is defined as

$$RUL(c_0) = EOP - c_0. \tag{46}$$

In practice, the predicted RUL is obtained by first determining the predicted EOP from the model's capacity forecast $\widehat{C}(\cdot)$:

$$\widehat{EOP} = \min\{c' \geq c_0 | \widehat{C}(c') \leq \gamma C_0\}, \quad \widehat{RUL}(c_0) = \widehat{EOP} - c_0. \tag{47}$$

Equations above indicate that, at a given evaluation point c_0, the total life can be expressed as $EOP = c_0 + RUL(c_0)$ for the true value and $\widehat{EOP} = c_0 + \widehat{RUL}(c_0)$ for the predicted value.

3.4. Experimental datasets

To assess the generalization ability and performance superiority of the proposed model, three battery datasets, Oxford, NASA, and CALCE, are selected for evaluation. These datasets comprise batteries with different capacities, chemistries, and charge–discharge protocols, and the cycle counts differ substantially between the two battery types. The main characteristics of these datasets are summarized in Table 3.

The Oxford battery dataset [55] contains data from eight Kokam SLPB533459H4 lithium-ion pouch cells, each featuring a nominal capacity of 740 mAh. These cells utilize lithium cobalt oxide (LCO) as the cathode material. Aging tests were conducted in a thermal chamber maintained at 40 °C. The charging protocol involved a constant current (CC) phase at 1C rate until the voltage reached 4.2 V, followed by a constant voltage (CV) phase at 4.2 V until the current tapered to 0.02C. Subsequently, the batteries were discharged following a drive cycle derived from the urban Artemis profile. Characterization measurements were taken every 100 cycles to monitor degradation trends. The eight cells capacity degradation curves are shown in Fig. 4(a)-(b).

Table 3
Basic information of each battery dataset.

Data Sources	Oxford [28]	NASA [15]	CALCE [29]
Cell types	Pouch	18650 Cylindrical	Prismatic
Cathode material	NCO-LCO	NCA	LCO
Selected battery series	Cell (1 to 8)	B00 (05 to 07 and 18)	CS2 (35 to 38)
Battery number	8	4	4
Rated capacity (Ah)	0.74	2	1.1
Capacity failure threshold (Ah)	70%	1.4 (70%)	0.84 (76.4%)
Charge protocol (rate)	CC (2C)	CC-CV (0.75C,4.2V)	CC-CV (0.5C,4.2V)
Discharge protocol (rate)	variance	CC (1C)	CC (1C)
Cut-off voltage (V)	charge to 4.2 discharge to 2.7	charge to 4.2 discharge to 2.7/ 2.5/2.2/2.5	charge to 4.2 discharge to 2.7
Cut-off current (A)	–	charge to 0.02	charge to 0.05
Experiment temperature (°C)	40	24	24
Cycle numbers	8200/7700/8100/ 5100/5000/5000/ 8100/8100	168/168/168/ 132	881/935/ 971/995

The NASA Prognostics Data Repository contains experimental data from four 18,650-model Li-ion batteries (B0005, B0006, B0007, and B0018) subjected to standardized aging tests under controlled ambient conditions (24 °C) [18]. Each battery underwent repeated charge-discharge cycles and impedance measurements to study degradation patterns. Charging followed a two-phase protocol: Constant Current (CC) at 1.5 A until reaching 4.2 V, followed by Constant Voltage (CV) until the current dropped to 20 mA. Discharge cycles were conducted at 2 A CC mode, with cutoff voltages varying by battery: 2.7 V (B0005), 2.2 V (B0007), and 2.5 V (B0006, B0018). The capacity degradation curves of these four batteries are shown in Fig. 4(c)-(d).

The CALCE battery datasets [56] includes four prismatic lithium-ion batteries manufactured by LG Chem, each with a nominal capacity of 1.1 Ah and employing lithium cobalt oxide as the cathode material. All aging experiments are conducted under room temperature conditions. The charging process follows a constant current protocol at 0.5 C until the voltage reaches 4.2 V, after which a constant voltage phase is applied at 4.2 V until the charging current falls below 0.02 A. Discharging is performed at a constant current of 2 A until the voltage drops to 2.7 V. The degradation patterns observed in the CALCE dataset are illustrated in Fig. 4(e).

3.5. Feature engineering

3.5.1. Health indicators extraction

In this section, voltage, temperature, time, and capacity data from the battery charging process are analyzed and processed to derive four characteristic curves: Incremental Capacity (IC), Charge Voltage Time (CVT), Differential Temperature Voltage (DTV), and Differential Temperature Capacity (DTC), as shown in Fig. 5(a)-(d). From these curves, ten representative HIs are extracted, capturing key features related to battery degradation. The ten HIs and their corresponding descriptions are summarized in Table 4.

Incremental Capacity (IC) analysis is a diagnostic technique used to assess the health of lithium-ion batteries by examining the differential relationship between capacity and voltage during charging or discharging processes [19]. The IC curve is obtained by computing the derivative of capacity (Q) with respect to voltage (V), represented mathematically as:

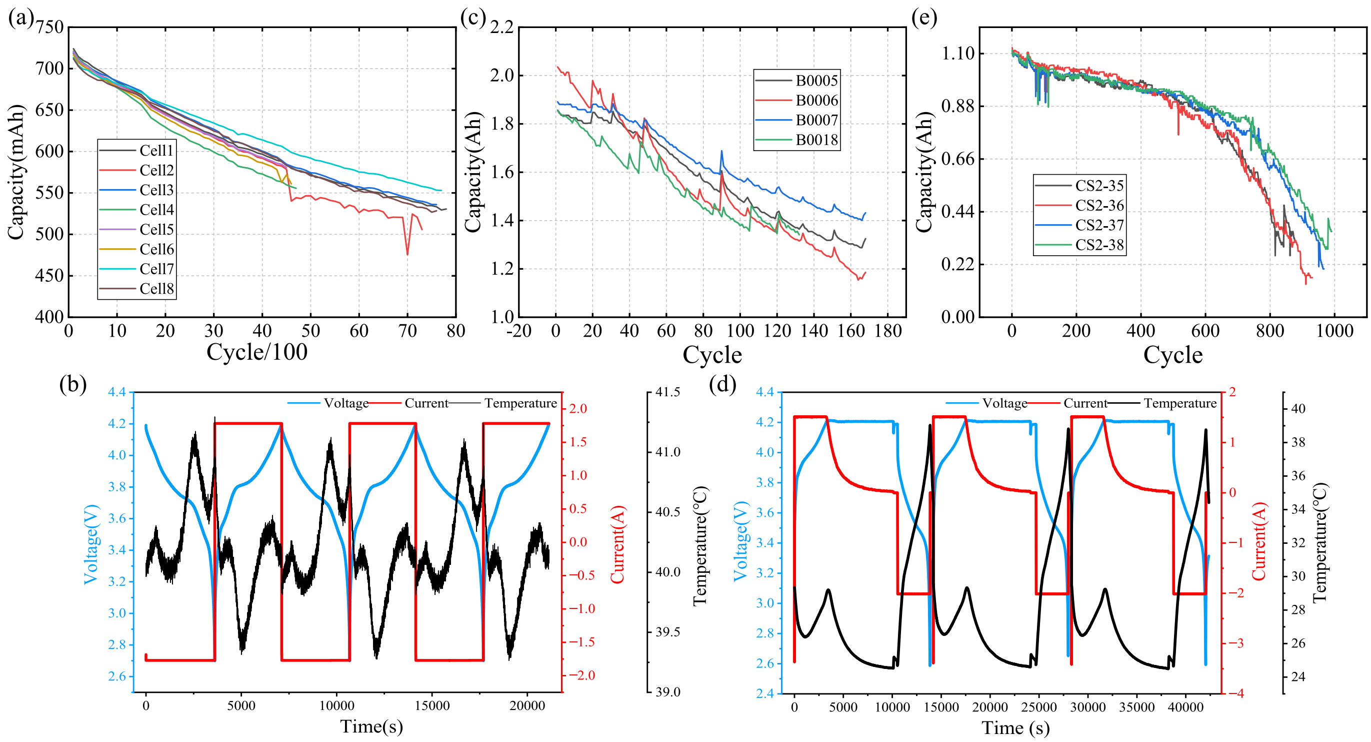

Fig. 4. SOH degradation curves and corresponding voltage, current, and temperature profiles of (a–b) Oxford, (c–d) NASA, and (e) CALCE datasets.

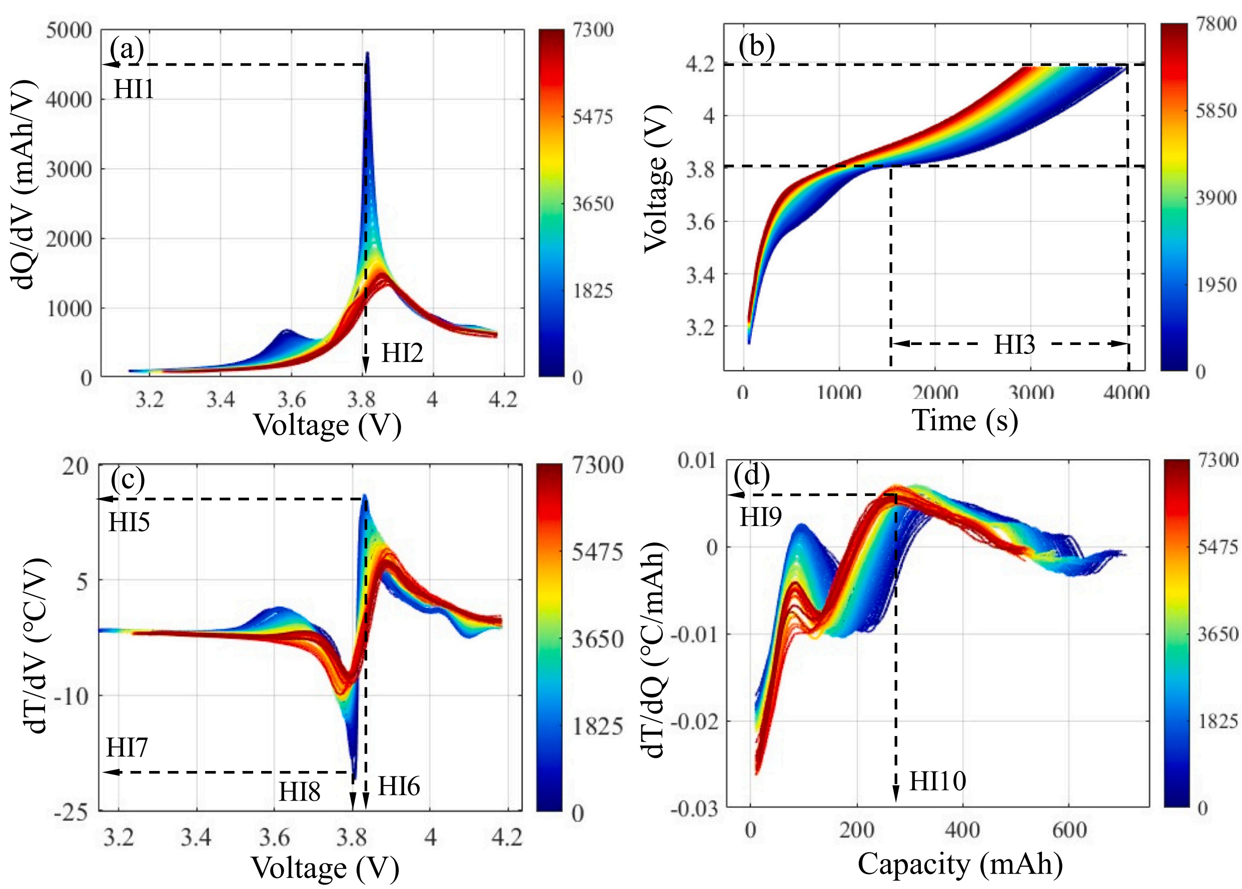

Fig. 5. All cycle curves of (a) DTV, (b) DTC, (c) IC and (d) CVT and the position of each HIs extraction.

Table 4
Multi-source health indicators.

No.	Description on Health Indicators
HI1	The peak value of the incremental capacity (IC) curves
HI2	The voltage corresponding to the peak of the incremental capacity (IC) curves
HI3	The charge time within the voltage range of 3.8–4.2 V
HI4	The charge time within the voltage range of 4.16–4.17 V
HI5	The peak value of the differential temperature–voltage (DTV) curves
HI6	The voltage corresponding to the peak of the differential temperature–voltage (DTV) curves
HI7	The valley value of the differential temperature–voltage (DTV) curves
HI8	The voltage corresponding to the valley of the differential temperature–voltage (DTV) curves
HI9	The peak value of the differential temperature–capacity (DTC) curves
HI10	The capacity corresponding to the peak of the differential temperature–capacity (DTC) curves

$$\mathrm{IC} = d\left(\frac{Q}{V}\right) = \frac{dQ}{dV} \approx \frac{\Delta Q}{\Delta V} \tag{48}$$

The IC curve typically exhibits distinct peaks corresponding to specific electrochemical reactions within the battery [57]. To better evaluate the role of the IC curve in revealing battery degradation information, the IC curve is analyzed during the charging process, as shown in Fig. 5(a), the peaks and their associated voltages are marked as HI1 and HI2, respectively. The charge voltage curve records the variation of battery voltage over time and is commonly used to extract HIs, such as the charging voltage time from 3.8 V to 4.2 V. Due to its simplicity in extraction and high correlation, inspired by a series of studies [2,20,52,58], the time intervals during which the charging voltage ranges from 3.8 V to 4.2 V and from 4.16 V to 4.17 V as health indicators are extracted as HI3 and HI4, respectively, as shown in Fig. 5 (b). The calculation formula is as follows:

$$\mathrm{CVT} = t(V_1) - t(V_2) \tag{49}$$

where t represents the charging time when the voltage reaches V_1 and V_2.

The Differential Temperature (DT) curve is a valuable diagnostic tool for assessing the SOH of lithium-ion batteries [59]. By plotting the rate of temperature change against voltage during charging or discharging, the DT curve highlights subtle thermal behaviors that are often imperceptible in standard temperature profiles. This enhanced sensitivity enables the detection of minor variations in internal resistance and heat generation, which are indicative of aging mechanisms such as electrode degradation or electrolyte decomposition. To construct the DTV curve, one calculates the derivative of temperature with respect to voltage, expressed mathematically as:

$$\mathrm{DTV} = d\left(\frac{T}{V}\right) = \frac{dT}{dV} \approx \frac{T(k+N) - T(k)}{V(k+N) - V(k)} \tag{50}$$

where T, V, k and N denotes the temperature, voltage, the k sample point and the interval of two sample points. Since the DTV curve contains significant noise, a moving average smoothing process with a window size of 200 is employed to mitigate this effect.

Considering that the peaks and valleys of the DTV curve can characterize the battery's thermal and electrochemical behaviors during charging, four HIs are extracted: the peak of the DTV curve (HI5), the voltage corresponding to the peak (HI6), the valley (HI7), and the voltage corresponding to the valley (HI8). These HIs reflect heat generation, internal resistance changes, and phase transitions within the battery, providing insights into its degradation characteristics. The processed DTV curve and the positions of these four HIs are shown in Fig. 5(c).

The DTC curve effectively highlights thermal anomalies, phase transitions, and heat generation patterns, making it a valuable indicator of battery aging and degradation. To leverage these characteristics, we extract the peak value of the DTC curve as HI9 and the corresponding capacity as HI10. The processed DTC curve and the positions of these extracted HIs are shown in Fig. 5(d). The DTC curve represents the rate of temperature change with respect to capacity during the charging process, providing insights into the battery's thermal characteristics and internal reaction mechanisms. It is defined as:

$$\mathrm{DTC} = d\left(\frac{T}{Q}\right) = \frac{dT}{dQ} \approx \frac{T(k+N) - T(k)}{Q(k+N) - Q(k)} \tag{51}$$

where T, Q, k and N denotes the temperature, capacity, the k sample point and the interval of two sample points. To reduce the impact of significant noise in the DTC curve, a moving average smoothing process with a window size of 200 is applied.

3.5.2. Health indicators selection

The extraction of HIs is a crucial step in the SOH estimation of lithium-ion batteries [60–62], as the quality of the extracted indicators directly determines the precise of the final SOH estimation. The Pearson (γ) correlation coefficient (PCC) and the Spearman (ρ) correlation coefficient (SCC) are calculated to analyze the correlation between HIs and SOH. PCC measures the linear correlation, while SCC assesses the monotonic relationship between variables, making it more suitable for capturing nonlinear dependencies. The formulas for PCC and SCC are given as follows:

$$\gamma = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum_{k}^{n} (X_k - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{k}^{n} (X_k - \overline{X})^2 \sum_{k}^{n} (Y_k - \overline{Y})^2}}, \tag{52}$$

where n,X_k,Y_k,$\overline{X}$ and $\overline{Y}$ are the number of sample, the value of true SOH, the HI, the mean of X and Y, respectively.

$$\rho = \frac{\frac{1}{n}\sum_{i=1}^{n}(R(X_i) - \overline{R(X)})(R(Y_i) - \overline{R(Y)})}{\sqrt{\left(\frac{1}{n}\sum_{i=1}^{n}(R(X_i) - \overline{R(X)})^2\right)\left(\frac{1}{n}\sum_{i=1}^{n}(R(Y_i) - \overline{R(Y)})^2\right)}}, \tag{53}$$

where $R(X_i)$ and $R(Y_i)$ are the ranks of the SOH and HI, respectively; n is the sample size.

Furthermore, the correlation coefficients for all HIs of each battery cell can be calculated using the above formulas. The heatmap of PCC and SCC for cell1 to cell4 is shown in Fig. 6, where the strength of correlation is visually represented using circle size and color intensity. Larger and darker circles indicate a stronger correlation between the HI and SOH, while smaller and lighter circles indicate weaker correlations. An HI with a strong correlation (greater than 0.9) in one battery may exhibit a weaker correlation (less than 0.9) in another. For example, the PCC and SCC of HI10 are 0.95 and 0.94 for Cell1, respectively, whereas in Cell 3, they decrease to 0.88 and 0.86. This discrepancy may affect the prediction accuracy of a model trained on Cell 1 when applied to Cell 3. Therefore, an effective method for selecting strongly correlated HIs that are applicable across multiple battery cells is essential.

Many existing SOH estimation methods select HIs solely based on the magnitude of their correlation coefficients. However, this approach often proves ineffective when the correlation distributions of different HIs vary significantly, such as under different battery chemistries or various temperature conditions. In these scenarios, an HI that performs well in one setting may not be suitable in another. Therefore, to ensure the robustness and applicability of selected HIs across diverse conditions, it is essential to introduce a more effective selection strategy that not only considers correlation coefficients but also incorporates a rational screening mechanism based on their distribution characteristics.

Therefore, a HI selection algorithm is proposed. This method first

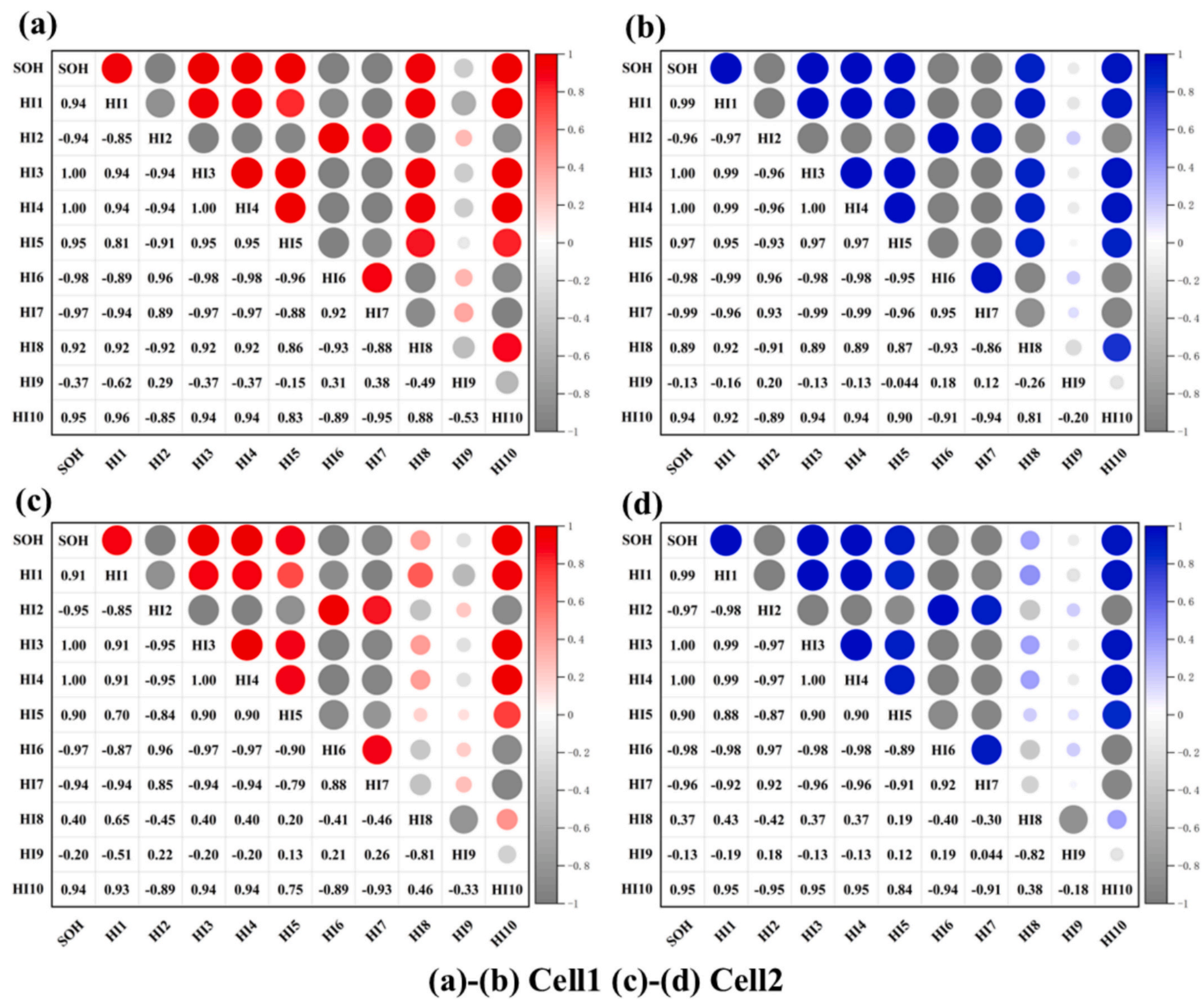

Fig. 6. Hotspot maps of PCC and SCC between SOH and ten HIs.

calculates both the Pearson and Spearman correlation coefficients of multiple candidate HIs across all battery cells to jointly evaluate their linear and monotonic nonlinear relationships with battery degradation. Only HIs whose Pearson and Spearman correlation coefficients are both greater than 0.8 are retained, ensuring that the selected HIs exhibit strong and robust relevance to SOH under different operating conditions. The retained HIs are then ranked according to their correlation strength, and features with inter-feature correlation coefficients higher than 0.95 are further removed to eliminate highly redundant information while preserving sufficient feature diversity. Based on this process, the final HI subset is obtained following the screening procedure described in Table 5.

Table 5
Steps of health indicators selection for a group of cells.

Procedure: HIs selection

1.Calculate the PCC (γ) and SCC (ρ) of n HIs for m cells
2.Obtain the average PCC ($\overline{\gamma_i}$) and SCC ($\overline{\rho_i}$) for each HI
3.Obtain i_{th} HI (f_i) with $\overline{\gamma_i} >= 0.8$ or $\overline{\rho_i} >= 0.8$, and order from highest to lowest
4.Construct $A = \left[f_1, f_2, \dots f_n\right]$
5.For $i = 1, 2, \dots, n$
For $j = 1, 2, \dots, m$
  If $\gamma_{ij} \leq 0.95$ or $\rho_{ij} \leq 0.95$
  Delete f_i from A
  End
 End
 End
 6.Output A

Through our HIs selection method, three HIs (HI3, HI4, and HI6) that meet the selection criteria are identified. The PCC and SCC values of these three selected HIs for each battery are shown in Table 6**.** It can be observed that both HI3 and HI4 exhibit strong correlations, with PCC and SCC values exceeding 0.99, indicating their consistently high performance across all batteries. In comparison, HI6 shows a slightly lower correlation, with an average value of 0.9789, but still meets the selection criteria.

The correlation-based HI selection is intentionally performed using data from multiple cells within the same battery group. This approach aims to ensure the statistical robustness and generalizability of the selected indicator, as a HI derived from a single cell may not adequately represent population-level degradation patterns. By evaluating candidates across several cells, we identify a composite HI that captures common degradation characteristics, thereby reducing reliance on cell-specific randomness.

Crucially, this selection process is confined to the within-group design phase. Once the optimal composite HI is determined for a given group, it is fixed and directly applied for feature extraction in subsequent health estimation tasks—including those involving unseen battery groups. No retrospective correlation analysis or re-selection is performed using data from the test sets. Therefore, the integrity of the evaluation is maintained, as the model training and testing phases strictly adhere to the principle of using mutually exclusive data sources.

4. Experimental results analysis

This study conducts a series of experiments to comprehensively evaluate the performance and robustness of the proposed model. Three types of experiments are carried out across three distinct datasets: SOH estimation, RUL prediction, and ablation and complexity analysis. The SOH experiments not only assess the predictive accuracy of the proposed model but also serve to validate the effectiveness of the selected health indicators, demonstrating their capability to capture critical degradation patterns. The RUL experiments further examine the model's ability to generalize across different battery types and operating conditions, providing a rigorous assessment of its prognostic reliability. Finally, ablation studies and computational complexity analyses are conducted to quantify the contributions of each module within the proposed architecture and to evaluate its efficiency in terms of model size and inference cost. Collectively, these experiments provide a thorough validation of both the accuracy and practical applicability of the proposed approach for lithium-ion battery health monitoring and remaining useful life prediction.

4.1. Experiments I: Comparative study on different HIs

To validate the effectiveness of the proposed method for selecting strongly correlated HIs applicable to all cells in a group, The first 30% data of Cell1 is chosen as the training data. Only the strongly correlated indicators of Cell1 (with absolute PCC and SCC values greater than 0.94) are considered, without considering the correlation strength of health indicators from other test batteries. To avoid redundancy, only one feature is selected from each type of curve. The final selected health indicators are HI1, HI4, HI6, and HI10, which are respectively abbreviated as the 'IC' feature, 'CVT' feature, 'DTV' feature, and 'DTC' feature. Additionally, all four HIs are fed into the model together for a comprehensive performance evaluation, referred as 'Fusion'. The SOH estimation results of each selected HI using the proposed model are shown in the Fig. 7.

Table 7 shows the MAE, RMSE, and MAPE metrics of different HIs for this group of batteries using the proposed model, as well as the average MAE and RMSE for multiple batteries on a single feature. The results indicate that using the CVT feature consistently achieves high estimation accuracy, with all experimental MAPE values below 1%, while the other three individual features yield relatively average performance. Notably, when all four HIs (Fusion) are used together as input, the final estimation accuracy ranks second in terms of average RMSE and third in terms of average MAE and MAPE, demonstrating that the overall correlation of health indicators also influences the final estimation accuracy.

Table 6
Selected HIs with PCC and SCC of all cells.

Battery	HI3 γ	HI3 ρ	HI4 γ	HI4 ρ	HI6 γ	HI6 ρ
Cell1	0.9999	0.9999	0.9999	0.9998	−0.9783	−0.9779
Cell2	0.9999	0.9998	0.9999	0.9999	−0.9731	−0.9785
Cell3	0.9999	0.9999	0.9999	0.9999	−0.9556	−0.9593
Cell4	0.9997	1.0000	0.9999	0.9994	−0.9699	−0.9843
Cell5	0.9998	1.0000	0.9998	0.9999	−0.9739	−0.9788
Cell6	0.9999	1.0000	0.9999	1.0000	−0.9686	−0.9758
Cell7	0.9999	0.9999	0.9999	0.9999	−0.9753	−0.9880
Cell8	0.9999	0.9999	0.9999	0.9999	−0.9785	−0.9883

Furthermore, Table 7 also reveals that the representational ability of the same HI varies significantly across different batteries. For example, the DTC feature performs significantly better than the IC feature on Cell5, whereas the opposite is observed for Cell3. However, the CVT feature, extracted using the proposed strong correlation-based HI selection method for battery groups, consistently achieves the best performance across all batteries. Therefore, considering the correlation of health indicators across all battery data is essential, as it greatly contributes to enhancing the model's generalization and robustness.

4.2. Experiments II: Comparative study on SOH

This subsection first presents the SOH estimation performance using individual features versus fused features within the same model, aiming to validate the effectiveness of the optimized health indicator selection methods. Subsequently, the selected health indicators are employed to conduct model comparisons and evaluations across three battery datasets with different materials and operating conditions, fully demonstrating the superiority, effectiveness, and generalization capability of the proposed model.

In the SOH prediction experiments, the input health indicators are selected and constructed according to the characteristics of the corresponding battery datasets.

For the CALCE and NASA datasets, the SOH prediction experiments adopt HI3 as the sole input feature, following the feature selection algorithm described in Section 3.5. Compared with the Oxford dataset, these two datasets exhibit more pronounced short-term local fluctuations and larger sampling intervals, which limits the number of health indicators that simultaneously satisfy the strict selection criteria defined in Section 3.5. Under these conditions, HI3 remains the most stable and informative indicator, effectively characterizing the long-term capacity degradation trajectory and demonstrating strong correlation and robustness across multiple cells. Moreover, in order to ensure the use of a common health indicator across different datasets and enhance the generalizability of the proposed method, HI3 is consistently selected for the CALCE and NASA datasets.

For the Oxford dataset, a multidimensional health indicator set consisting of HI3, HI4, and HI6 is employed as the model input. These pre-selected HIs, identified using the same selection strategy, jointly provide richer feature representations, which are beneficial for capturing the more complex and smoother degradation behaviors observed in the Oxford dataset.

4.2.1. SOH estimation results on CS2 batteries

To verify the effectiveness of the proposed model across different electrode materials and charge–discharge rates, additional experiments were conducted on the CS2 batteries. In particular, the CS2–35 cell was employed for model training and validation, where the first 30% of the cycle data were used for training and the remaining 70% for validation. This setup ensures that the model is evaluated on unseen data from the same cell. Furthermore, to assess the generalization capability and robustness of the model, the trained network was tested on three additional cells under identical operating conditions. This experimental design enables comprehensive verification of both within-cell prediction accuracy and cross-cell generalization performance.

As illustrated in Fig. 8, the proposed model achieves the closest fit to the actual SOH values across the four CS2-series cells, which is of considerable importance for practical applications. Moreover, the boxplots in the upper-right corner reveal that the error distribution of the proposed model lies closest to the zero axis, indicating its superior predictive performance during the experiments. The detailed subfigures further demonstrate the model's ability to effectively capture short-term capacity fluctuations. These results highlight the strong predictive accuracy and generalization capability of the proposed model.

Table 8 further summarizes the predictive performance of the

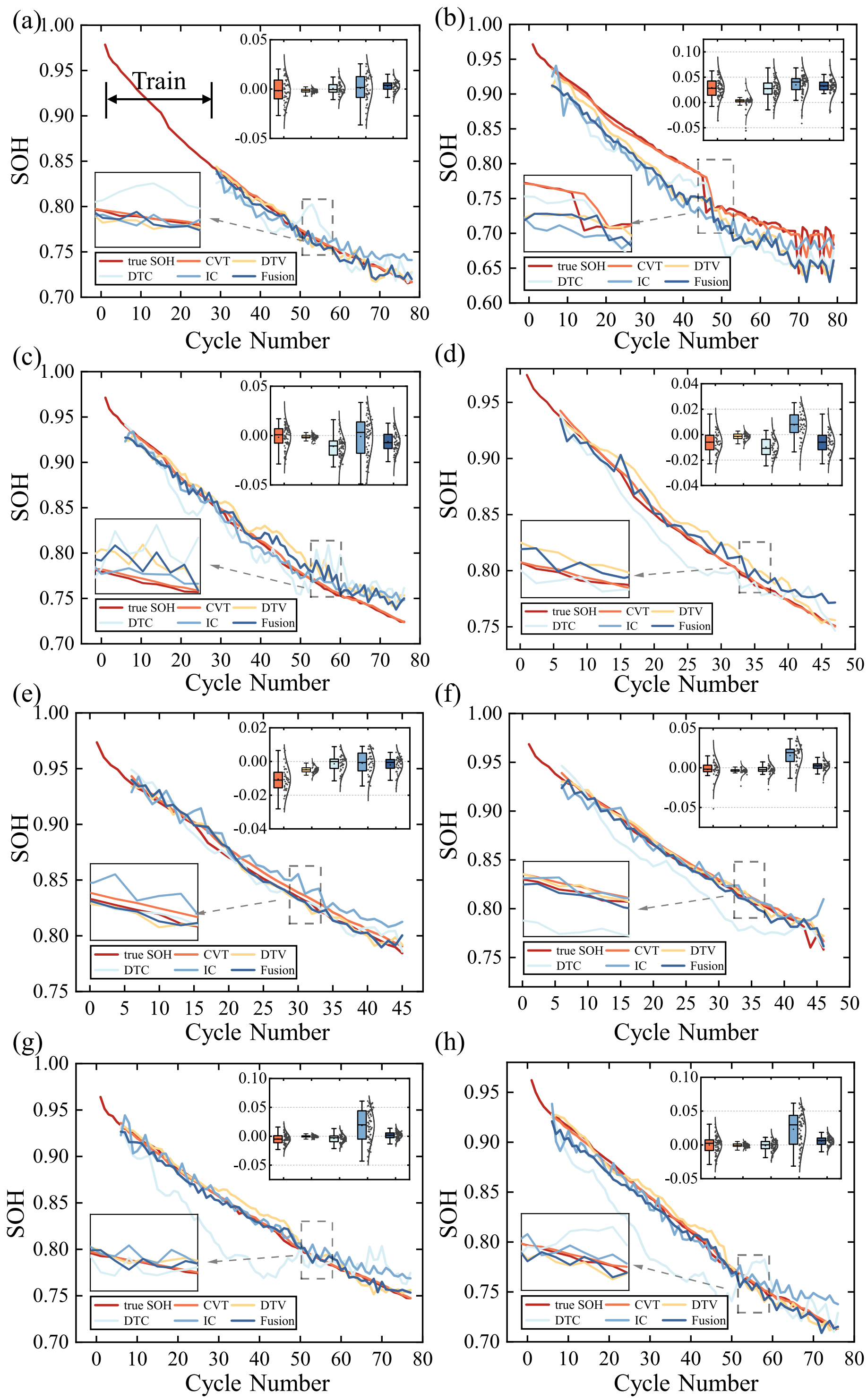

Fig. 7. SOH estimation results and error with different HIs on the Oxford dataset across (a) Cell 1 to (h) Cell 8.

proposed model and baseline methods across the four CS2-series cells. Overall, the proposed model achieves the lowest errors on all metrics, with average reductions of approximately 16.8% in MAE and 12.9% in RMSE compared with the best-performing baseline. For example, on CS2–35, the proposed method attains an MAE of 0.0141 and an RMSE of 0.0208, outperforming the CNN-Transformer with an MAE of 0.0147 and an RMSE of 0.0233, and substantially surpassing the conventional Transformer with an MAE of 0.0233 and an RMSE of 0.0294 as well as the MLP with an MAE of 0.0265 and an RMSE of 0.0416. A similar trend is observed across CS2–36, CS2–37, and CS2–38, further confirming the robustness and stability of the proposed architecture. In addition, although the CNN-Transformer exhibits relatively competitive performance on CS2–35 and CS2–36, its prediction accuracy degrades significantly on CS2–38, yielding a MAPE of 0.0528 compared to 0.0301 of the proposed model. This suggests that the CNN-Transformer is more sensitive to variations in cell characteristics, whereas the proposed model maintains stronger generalization ability. Finally, conventional deep learning models such as MLP, CNN, and LSTM show considerably higher prediction errors than attention-based approaches. In particular, CNN produces the largest errors across all cells, with an RMSE of 0.0643

Table 7
Three performance metrics for evaluating different HIs on the Oxford dataset.

Cell	MAE					RMSE					MAPE				
	IC	CVT	DTV	DTC	Fusion	IC	CVT	DTV	DTC	Fusion	IC	CVT	DTV	DTC	Fusion
Cell1	1.04%	0.19%	0.47%	1.27%	0.50%	1.26%	0.24%	0.63%	1.52%	0.60%	1.37%	0.24%	0.62%	1.64%	0.65%
Cell2	3.23%	0.65%	2.72%	3.75%	3.29%	3.59%	1.19%	3.08%	3.99%	3.46%	4.02%	0.87%	3.49%	4.78%	4.19%
Cell3	0.87%	0.19%	1.35%	1.65%	1.03%	1.09%	0.25%	1.57%	2.00%	1.23%	1.10%	0.24%	1.70%	2.08%	1.30%
Cell4	0.83%	0.24%	1.03%	1.13%	0.83%	1.01%	0.33%	1.21%	1.32%	1.01%	1.02%	0.28%	1.25%	1.36%	1.02%
Cell5	1.16%	0.48%	0.39%	0.56%	0.37%	1.30%	0.52%	0.52%	0.68%	0.53%	1.37%	0.57%	0.45%	0.65%	0.43%
Cell6	0.70%	0.40%	0.42%	1.82%	0.44%	1.14%	0.54%	0.62%	2.05%	0.58%	0.85%	0.49%	0.51%	2.16%	0.53%
Cell7	0.82%	0.15%	0.61%	2.80%	0.51%	1.03%	0.20%	0.79%	3.32%	0.61%	1.01%	0.18%	0.75%	3.35%	0.61%
Cell8	0.99%	0.23%	0.59%	2.95%	0.71%	1.25%	0.28%	0.76%	3.43%	0.85%	1.25%	0.28%	0.74%	3.57%	0.85%
Ave	1.21%	0.32%	0.95%	1.99%	0.96%	1.46%	0.44%	1.15%	2.29%	1.11%	1.50%	0.67%	1.19%	2.45%	1.20%

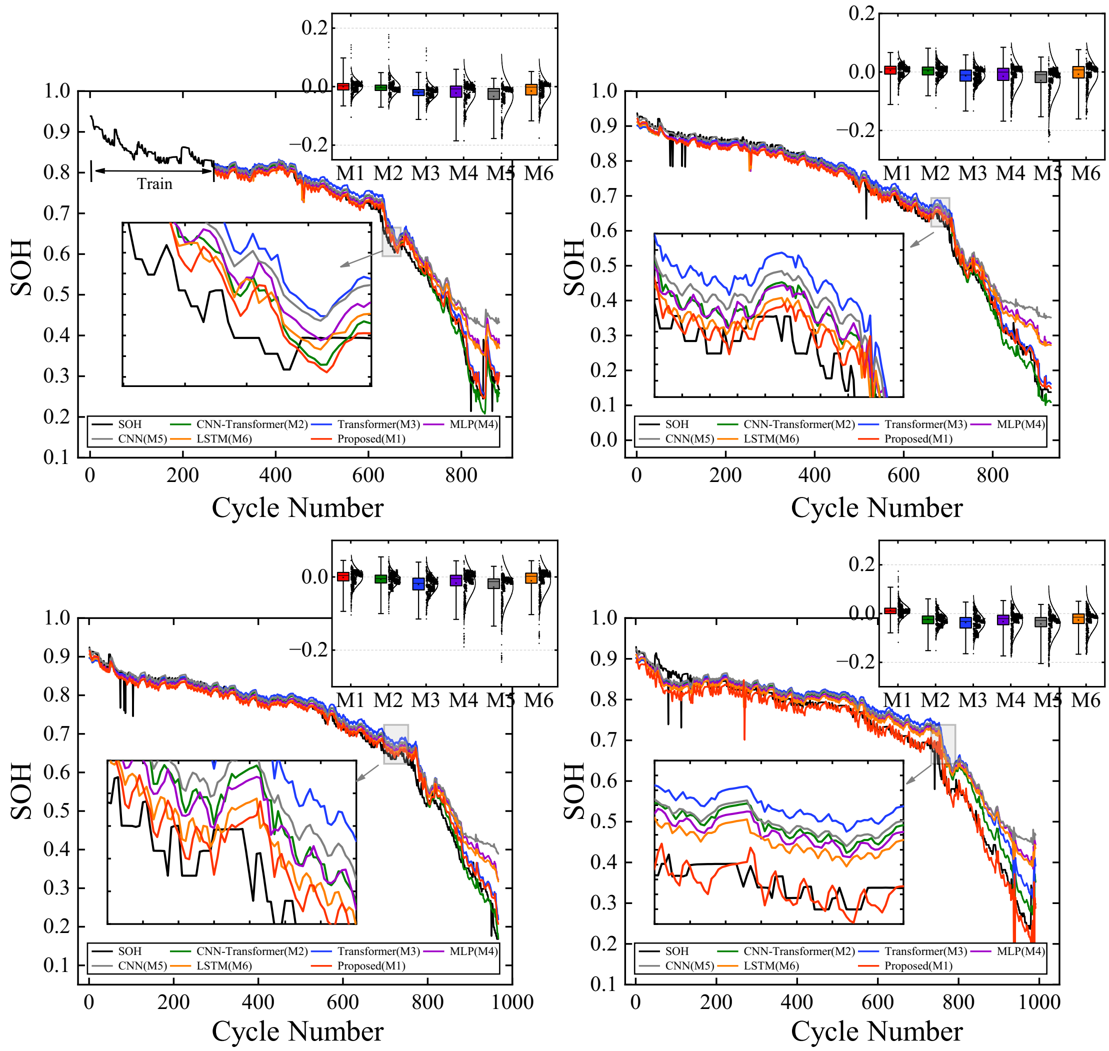

Fig. 8. The comparison of SOH estimation results on (a) CS2–35, (b) CS2–36, (c) CS2–37, and (d) CS2–38.

on CS2–38, underscoring its limited capability in capturing long-term dependencies. LSTM performs better than CNN and MLP but still lags behind Transformer-based architectures.

4.2.2. SOH estimation results on NASA batteries

To evaluate the model's generalization capability and performance across different operational conditions of battery data, we conducted comparative experiments on the NASA dataset.

As shown in Fig. 9, the proposed method achieves superior performance across all five experimental trials and multiple battery datasets, including B0005, B0006, B0007, and B0018. As shown in Table 9, for instance, on the B0005 dataset, the proposed method achieves an MAE of 0.0046, significantly lower than CNN-Transformer at 0.0061, Transformer at 0.0094, and CNN-LSTM at 0.0104. The LSTM model performs the worst with an MAE of 0.0165. Compared to traditional machine learning approaches such as Back Propagation (BP) neural networks and Extreme Learning Machine (ELM) [63], which report MAEs of 0.0269 and 0.0274 respectively, the proposed method reduces the error by an order of magnitude, highlighting its enhanced accuracy in battery SOH estimation.

Table. 8
SOH estimation results of different models on CS2 batteries.

Battery		CS2–35	CS2–36	CS2–37	CS2–38
MAE	Proposed	0.0141	0.0174	0.0155	0.0182
	CNN-Transformer	0.0147	0.0177	0.0158	0.0302
	Transformer	0.0233	0.0218	0.0234	0.0415
	MLP	0.0265	0.0286	0.0235	0.0393
	CNN	0.0349	0.0326	0.0287	0.0463
	LSTM	0.0238	0.0276	0.0219	0.0350
RMSE	Proposed	0.0208	0.0212	0.0209	0.0266
	CNN-Transformer	0.0233	0.0220	0.0218	0.0356
	Transformer	0.0294	0.0281	0.0310	0.0501
	MLP	0.0416	0.0436	0.0389	0.0558
	CNN	0.0559	0.0580	0.0482	0.0643
	LSTM	0.0370	0.0399	0.0350	0.0511
MAPE	Proposed	0.0273	0.0319	0.0299	0.0301
	CNN-Transformer	0.0300	0.0359	0.0308	0.0528
	Transformer	0.0439	0.0443	0.0453	0.0767
	MLP	0.0667	0.0889	0.0585	0.0841
	CNN	0.0897	0.1161	0.0728	0.0989
	LSTM	0.0594	0.0819	0.0531	0.0758

The RMSE results further confirm the robustness of the proposed method. On the B0006 dataset, it records an RMSE of 0.0130, while CNN-Transformer and Transformer yield higher errors of 0.0177 and 0.0269. In contrast, the Gaussian Process Regression (GPR) model [64] exhibits a substantially higher RMSE of 0.4129 on the same dataset, indicating that the proposed method provides more stable and precise predictions.

Regarding MAPE, the proposed method maintains errors below 1.1% across all datasets, with the lowest being 0.63% on B0005. In comparison, Transformer reaches 3.28% on B0006, and LSTM reaches 2.40% on B0018. The BP and ELM models report MAPEs of 2.58% and 3.74% on B0006, while the GPR model from reference 40 shows an even higher error of 5.80% on B0018. These results underscore the proposed

Table. 9
SOH estimation results of different models on NASA batteries.

Battery		B0005	B0006	B0007	B0018
MAE	Proposed	0.0046	0.0072	0.0076	0.0082
	CNN-Transformer	0.0061	0.0106	0.0079	0.0106
	Transformer	0.0094	0.0240	0.0100	0.0126
	CNN-LSTM	0.0104	0.0158	0.0086	0.0122
	LSTM	0.0165	0.0159	0.0138	0.0181
	BP [63]	0.0269	0.0559	0.0498	0.0460
	ELM [63]	0.0274	0.1045	0.0532	0.0404
	GPR [64]	0.1896	0.2969	0.3992	0.4403
RMSE	Proposed	0.0067	0.0130	0.0102	0.0121
	CNN-Transformer	0.0078	0.0177	0.0093	0.0140
	Transformer	0.0110	0.0269	0.0120	0.0163
	CNN-LSTM	0.0121	0.0198	0.0106	0.0147
	LSTM	0.0181	0.0203	0.0155	0.0198
	BP [63]	0.0203	0.0412	0.0230	0.0343
	ELM [63]	0.0337	0.0632	0.0349	0.0314
	GPR [64]	0.2148	0.4129	0.7400	0.4761
	SVR [65]	0.0075	0.0166	0.0097	–
MAPE	Proposed	0.0063	0.0090	0.0093	0.0099
	CNN-Transformer	0.0083	0.0125	0.0093	0.0136
	Transformer	0.0126	0.0328	0.0123	0.0159
	CNN-LSTM	0.0128	0.0192	0.0103	0.0151
	LSTM	0.0233	0.0212	0.0176	0.024
	BP [63]	0.0227	0.0258	0.0126	0.0196
	ELM [63]	0.0221	0.0374	0.0187	0.0196
	GPR [64]	0.0239	0.0481	0.0491	0.0580

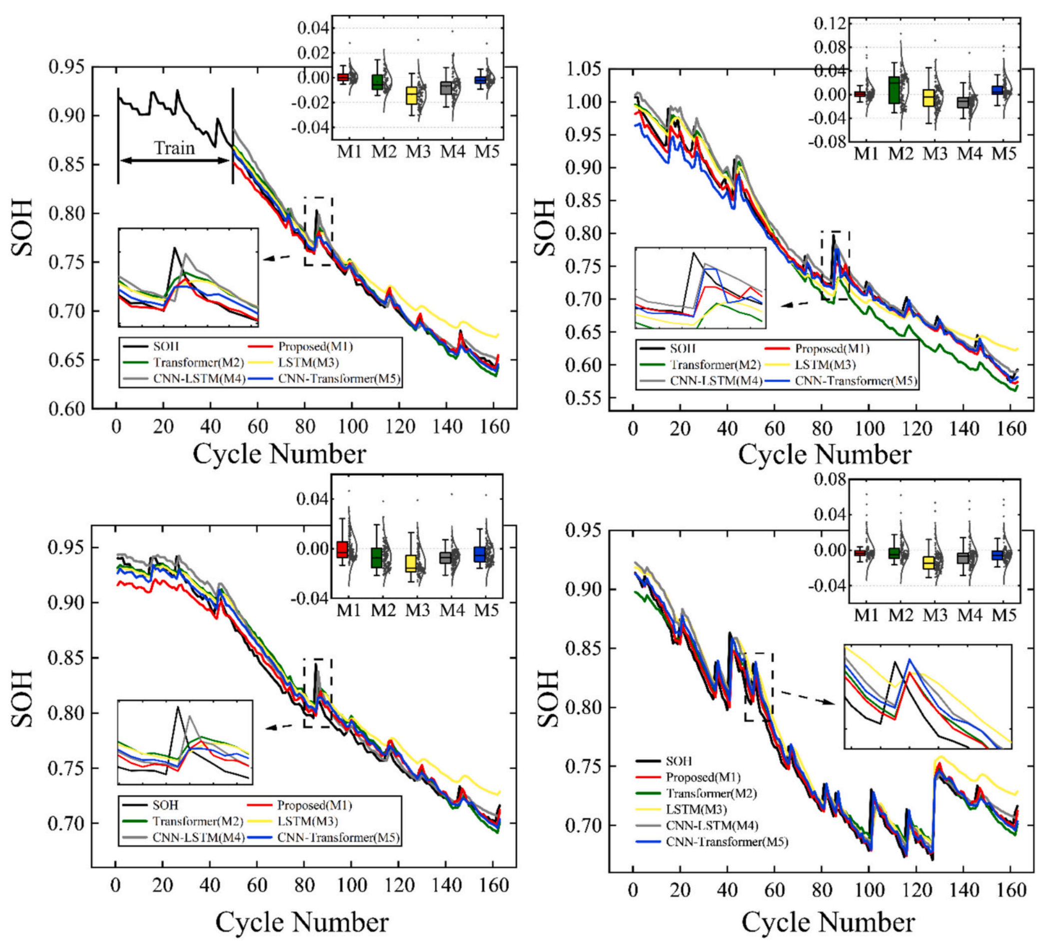

Fig. 9. The comparison of SOH estimation results on NASA batteries.

method's exceptional predictive capability.

In summary, the proposed method demonstrates strong robustness and generalization in battery state-of-health prediction, consistently outperforming both deep learning models and traditional machine learning techniques. Its low error rates and high reliability make it a valuable tool for optimizing battery management systems.

4.2.3. SOH estimation results on Oxford batteries

To validate the effectiveness and stability of the proposed model, the widely used LSTM model, the Transformer model, and the CNN-Transformer model, which incorporates convolutional layers, were selected as benchmark models for comparison. The SOH estimation results of the five models are presented in Fig. 10. Overall, the proposed model achieves the closest match to the actual SOH values across all cells, demonstrating superior validation and test performance. Even for Cell2, where the degradation trend is highly unstable, the proposed model exhibits the best fit, with the absolute error remaining closest to the zero axis.

In contrast, although the Transformer model performs well on Cell1, achieving results comparable to the proposed model, its estimation

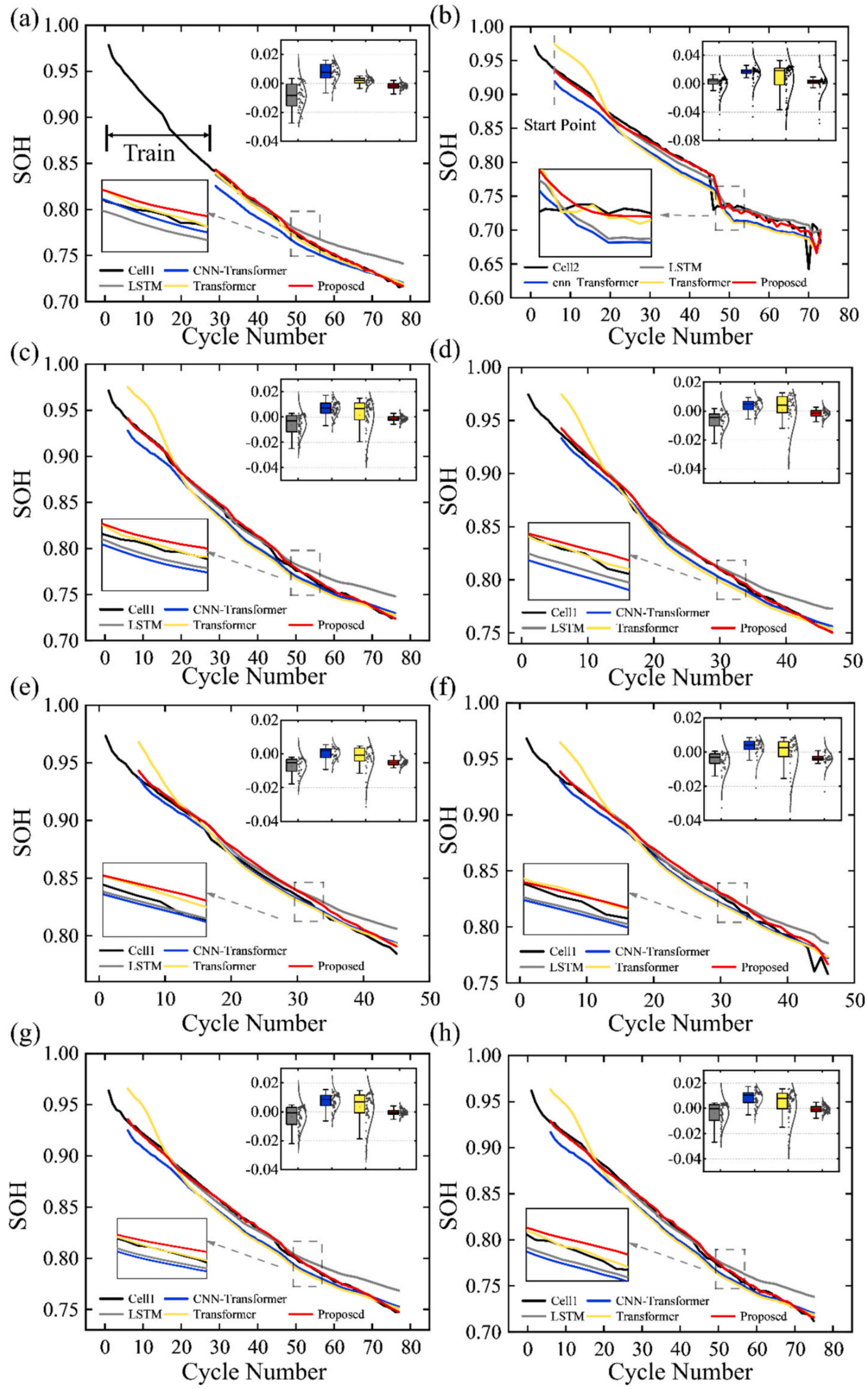

Fig. 10. SOH estimation results comparison with different models on the Oxford dataset across (a) Cell 1 to (h) Cell8.

accuracy deteriorates significantly on other cells, indicating overfitting. This issue arises due to the Transformer's weak ability to extract local features, making it less effective in distinguishing different battery degradation patterns. While the CNN-Transformer model improves upon this aspect, its overall accuracy remains far below that of the proposed model. Similarly, the estimation results reveal the limited ability of LSTM model to capture local features. Its estimated SOH curves appear overly smooth, allowing only a rough approximation of the true SOH values without accurately identifying short-term capacity growth phenomena in batteries.

Table 10 displays the estimation inaccuracies of eight cells for five models. In general, the proposed model shows higher estimation accuracy and achieves the smallest average estimation error. Specifically, the maximum estimation error of the proposed model remains within 1%, with a maximum MAE of 0.65% and a maximum RMSE of 0.87%, while the average MAE and RMSE are 0.32% and 0.40%, respectively. These outcomes suggest a more consistent performance in comparison to the other models, which is essential for maintaining the safe and reliable operation of batteries when handling substantial amounts of unseen data.

Furthermore, Table 10 also reveals that although the Transformer model performs not too bad on the training data, with results only slightly inferior to the proposed model, it suffers from overfitting. This is evidenced by its significantly poorer performance on the other test data, whereas the proposed model effectively mitigates this issue, leading to better overall results. This suggests that our model learns genuinely useful information from health indicators rather than merely fitting the degradation trend of the current battery. In contrast, while the LSTM model exhibits better generalization than the Transformer model, its learning capability on training data is noticeably weaker. Across various battery datasets, the proposed model outperforms it consistently.

4.3. Experiments III: Comparative study on RUL

This section presents a comparative study of the proposed model and several benchmark methods on the Remaining Useful Life (RUL) prediction task. Two representative datasets are employed to comprehensively evaluate the generalization and robustness of each model under different data availability conditions. Specifically, experiments are conducted with 10% and 30% of the total data used for training, aiming to assess the models' capability to perform accurate RUL estimation in both data-limited and data-sufficient scenarios.

The prediction results are analyzed from both numerical and visual perspectives. Quantitative metrics, including Absolute Error (AE) and Relative Error (RE), are used to evaluate prediction accuracy, while graphical comparisons between predicted and actual degradation trajectories illustrate each model's ability to capture capacity fading patterns.

The following subsections detail the RUL prediction performance on the CALCE dataset (Section 4.2.1) and NASA dataset (Section 4.2.2), highlighting the advantages and limitations of different approaches.

4.3.1. RUL prediction results on CS2 batteries

As shown in Fig. 11 and Table 11, the proposed model demonstrates stable and accurate RUL prediction performance on the CS2 series batteries (CS2–35 to CS2–38). Across all samples and training ratios, the prediction trends closely follow the actual capacity degradation curves, while the overall magnitude of errors remains comparatively small. This suggests that the model maintains good adaptability to different degradation trajectories under varying data availability.

In Fig. 11**(a–f)**, the predicted capacity curves obtained by the proposed model are well aligned with the noise-reduced experimental data, particularly near the End-of-Performance (EoP) threshold (0.88 Ah) where degradation accelerates. In contrast, other baseline models such as CNN and LSTM exhibit noticeable deviations, either overestimating or underestimating capacity during the later cycles. The Transformer-based models (Transformer and CNN-Transformer) generally achieve more stable predictions than CNN and MLP, but still show minor deviations near EoP. The proposed model presents a smoother and more consistent trajectory, indicating its ability to capture both local and long-term degradation dynamics.

Quantitatively, Table 11 reports the absolute error (AE) and relative error (RE) for each cell under 10% and 30% training data. The proposed model achieves AE values generally within ± 10 cycles across all cases, while the RE remains below 0.07, and often under 0.02.

To further investigate cases with relatively high prediction errors, we analyzed the influence of model structure on performance. Simpler models, such as CNN and MLP, generally exhibit higher errors due to their limited learning capacity and weaker ability to capture complex degradation patterns. For example, in CS2–35, the RE values are 0.0035 (10%) and 0.0698 (30%), notably lower than those of CNN-Transformer (0.0227, 0.0209) and Transformer (0.0925, 0.0419). Similarly, for CS2–38, the proposed model attains an RE of 0.0215 and 0.0017, whereas CNN and MLP exceed 0.26. These results indicate that the proposed model effectively captures degradation trends even when trained on limited data, maintaining reliable performance across different cells.

Overall, this comparison highlights that the observed differences in prediction errors are closely related to the structural complexity and learning capacity of the models, emphasizing the advantage of the proposed architecture in handling both SOH and RUL estimation tasks.

4.3.2. RUL prediction results on NASA batteries

Fig. 12 illustrates the comparison of the predicted and reference capacity degradation trajectories for four batteries (B0005, B0006, B0007, and B0018) from the NASA dataset. The proposed model exhibits a strong consistency with the reference capacity curves across all cases, accurately capturing both the early-stage nonlinear decay and the late-stage degradation trends. The predicted EoP points are closely aligned with the actual threshold (1.6 Ah), indicating that the model effectively predicts the RUL without significant delay or premature estimation.

Quantitative comparisons are summarized in Table 12, where the AE and RE are evaluated under different training data ratios (10% and

Table 10
Estimation errors of SOH for different models on the Oxford dataset.

Cell	MAE				RMSE				MAPE			
	Ours	CNN-Transformer	Transformer	LSTM	Ours	CNN-Transformer	Transformer	LSTM	Ours	CNN-Transformer	Transformer	LSTM
Cell1	0.19%	0.83%	0.26%	0.97%	0.24%	0.97%	0.30%	1.23%	0.24%	1.06%	0.34%	1.34%
Cell2	0.65%	1.77%	2.14%	0.72%	0.87%	2.24%	2.31%	1.17%	0.87%	2.24%	2.67%	0.95%
Cell3	0.19%	0.78%	1.11%	0.70%	0.25%	0.89%	1.41%	0.99%	0.24%	0.93%	1.30%	0.91%
Cell4	0.24%	0.49%	0.94%	0.63%	0.33%	0.55%	1.29%	0.85%	0.28%	0.58%	1.09%	0.79%
Cell5	0.48%	0.31%	0.52%	0.72%	0.52%	0.35%	0.86%	0.87%	0.57%	0.36%	0.59%	0.86%
Cell6	0.40%	0.48%	0.78%	0.53%	0.54%	0.62%	0.90%	0.86%	0.49%	0.58%	0.90%	0.66%
Cell7	0.15%	0.82%	0.98%	0.61%	0.20%	0.91%	1.21%	0.83%	0.18%	0.97%	1.15%	0.78%
Cell8	0.23%	0.91%	1.11%	0.69%	0.28%	1.11%	1.37%	0.94%	0.28%	1.09%	1.32%	0.90%
Average	0.32%	0.80%	0.98%	0.70%	0.40%	0.96%	1.21%	0.97%	0.39%	0.98%	1.17%	0.90%

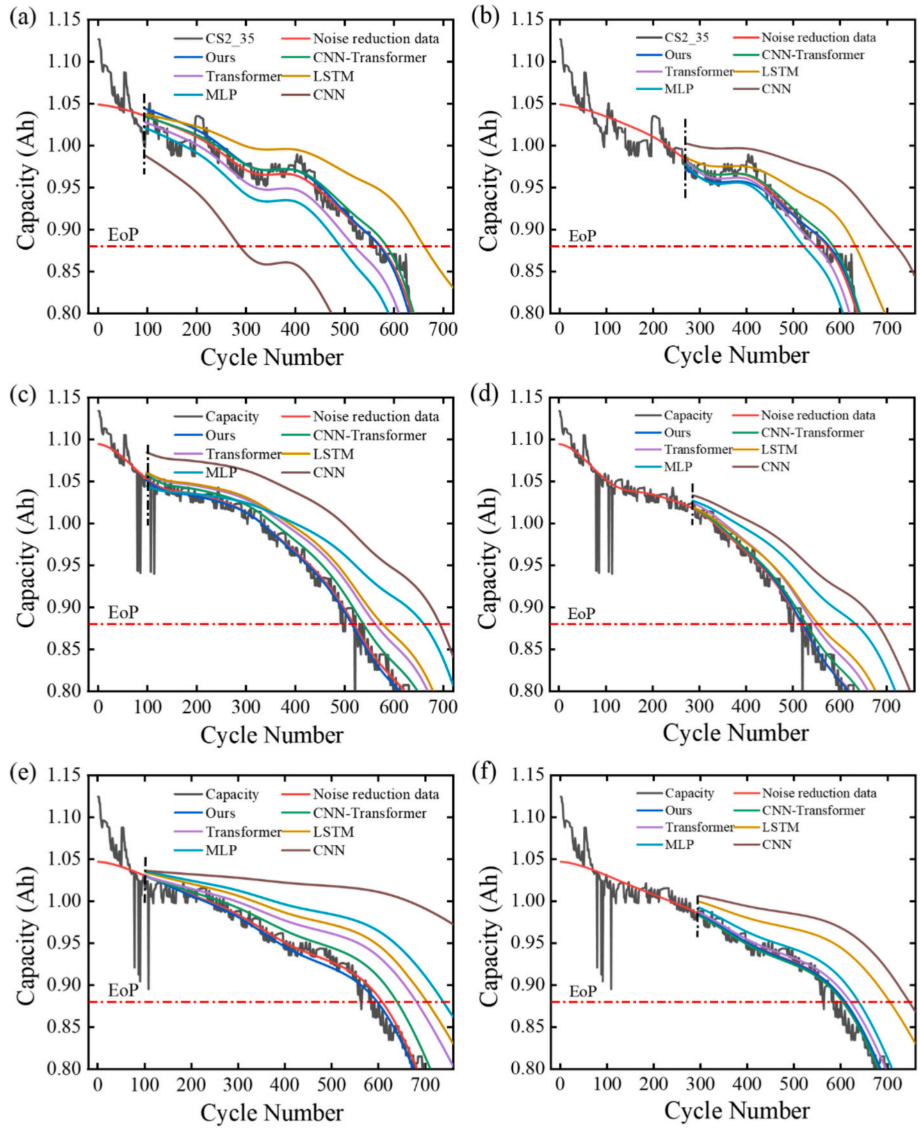

Fig. 11. RUL prediction results at early (10%) and mid-life (30%) stages for CS2–35 (a–b), CS2–36 (c–d), and CS2–37 (e–f).

Table. 11
RUL prediction results of different models on CALCE batteries.

Performance	Model	CS2–35		CS2–36		CS2–37		CS2–38	
		10%	30%	10%	30%	10%	30%	10%	30%
AE	CNN-Transformer	−13	−12	−19	−10	−31	4	−14	−3
	Transformer	53	24	−44	−27	−73	−12	−65	−15
	LSTM	−89	−61	−58	−36	−95	−97	−110	−43
	MLP	81	51	−142	−115	−130	−30	−147	−157
	CNN	285	−148	−175	−163	−364	−138	−170	−161
	Proposed	2	−4	4	−2	11	−1	−13	−1
RE	CNN-Transformer	0.0227	0.0209	0.0341	0.0180	0.0508	0.0066	0.0231	0.0049
	Transformer	0.0925	0.0419	0.0791	0.0486	0.1197	0.0197	0.1072	0.0248
	LSTM	0.1553	0.1065	0.1043	0.0647	0.1557	0.1590	0.1815	0.0710
	MLP	0.1414	0.0890	0.2554	0.2068	0.2131	0.0492	0.2426	0.2591
	CNN	0.4974	0.2583	0.3147	0.2932	0.5967	0.2262	0.2805	0.2657
	Proposed	0.0035	0.0698	0.0072	0.0036	0.0180	0.0016	0.0215	0.0017

30%). Across all battery samples, the proposed model achieves the lowest AE and RE values, demonstrating its better accuracy and generalization capability. Specifically, the average AE of the proposed method remains within ±5 cycles, significantly outperforming the CNN, LSTM, and Transformer-based models, which exhibit higher deviations—particularly for CNN and LSTM models that show substantial overestimation and underestimation in certain cases. Moreover, the proposed model consistently maintains RE values below 0.08, whereas other models frequently exceed 0.1 under the same conditions.

These results confirm that the proposed method can effectively

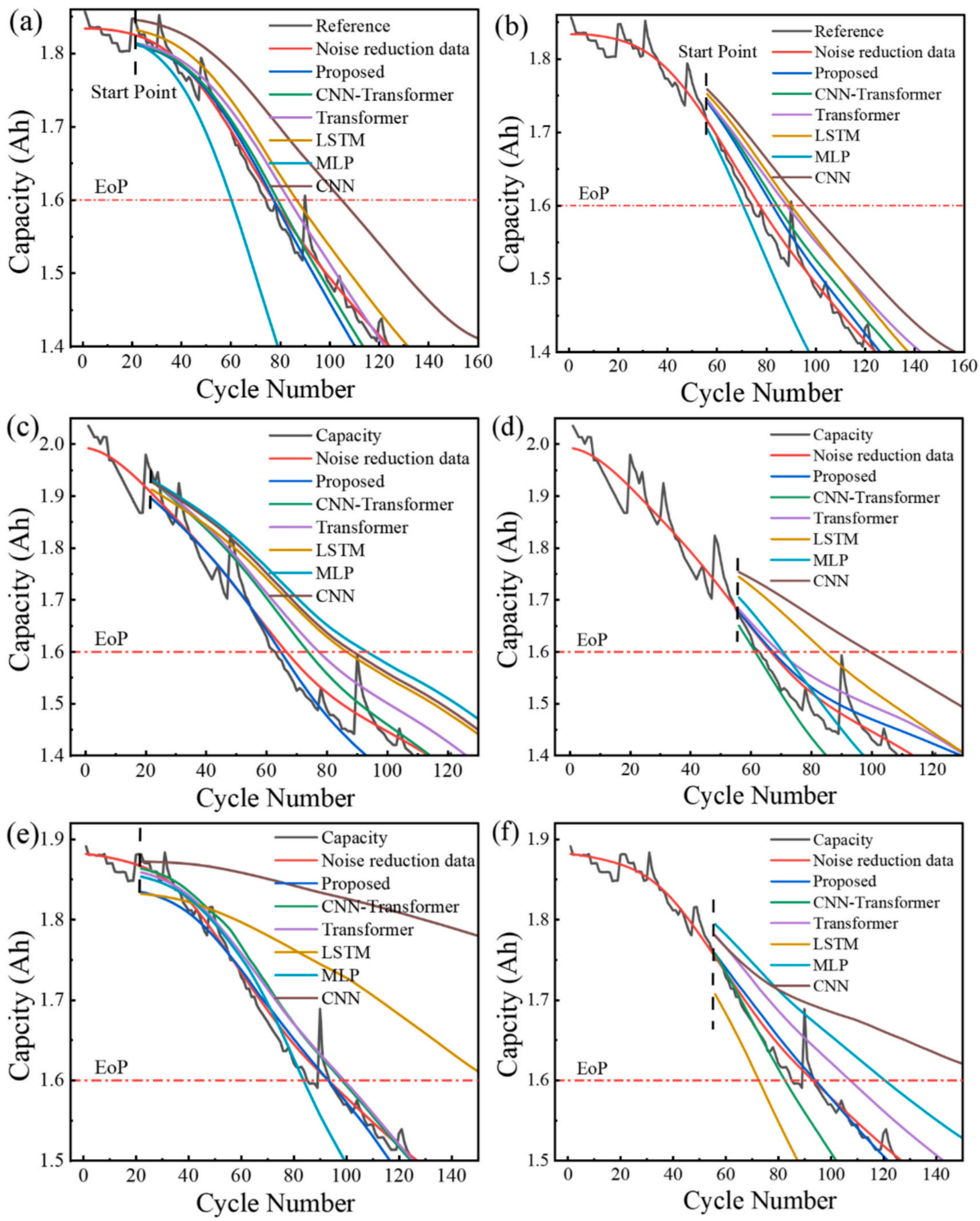

Fig. 12. RUL prediction results at early (10%) and mid-life (30%) stages for B0005 (a–b), B0006 (c–d), and B0007 (e–f).

capture the degradation patterns and nonlinear dynamics of lithium-ion batteries, ensuring precise RUL estimation even when limited historical data (e.g., 10% of life cycles) are available. This robustness highlights the model's potential for practical deployment in battery management systems (BMS) requiring real-time health monitoring and long-term prediction accuracy.

4.4. Experiments IV: Ablation studies and complexity analysis

To further validate the effectiveness and efficiency of the proposed model, additional experiments are conducted focusing on ablation studies and complexity analysis. The ablation study aims to clarify the individual contributions of key modules by systematically removing them and comparing the results, thereby highlighting the necessity of each component. In parallel, the complexity analysis provides a quantitative evaluation of the model's computational cost, ensuring that the performance gains are achieved without excessive overhead. Together, these experiments offer a comprehensive assessment of both predictive accuracy and practical feasibility.

4.4.1. Ablation study

Linear attention mechanisms significantly reduce the computational and memory complexity of standard attention by employing techniques such as low-rank decomposition [66,67], kernel approximation [68], sparsification [69], and local attention strategies [70,71]. While these approaches improve efficiency, they often compromise representation capacity and feature extraction ability to varying degrees. Conventional convolution operations can partially alleviate this limitation; however,

Table. 12
RUL prediction results of different models on NASA batteries.

Performance	Model	B0005		B0006		B0007		B0018	
		10%	30%	10%	30%	10%	30%	10%	30%
AE	CNN-Transformer	−1	−7	−7	5	−5	11	−9	−7
	Transformer	−6	−11	−11	−3	−6	−14	−15	−8
	LSTM	−9	−13	−20	−18	−63	21	−16	−9
	MLP	17	7	−27	−4	10	−27	−18	−9
	CNN	−27	−18	−22	−33	−75	−75	−77	−17
	Proposed	0	−5	2	−1	0	−1	2	−4
RE	CNN-Transformer	0.0128	0.0843	0.1077	0.0735	0.0532	0.1170	0.1607	0.1250
	Transformer	0.0769	0.1294	0.1486	0.0048	0.0064	0.1489	0.2679	0.1429
	LSTM	0.1154	0.1461	0.2564	0.2571	0.6702	0.2234	0.2857	0.1607
	MLP	0.2179	0.0769	0.3103	0.0471	0.1064	0.2872	0.3214	0.1607
	CNN	0.3462	0.2535	0.2340	0.4648	0.7979	0.7979	1.3750	0.3036
	Proposed	0	0.0641	0.0299	0.0149	0	0.0106	0.0357	0.0714

they usually introduce a substantial number of additional parameters, resulting in an unfavorable trade-off between accuracy and efficiency.

To address this issue, the proposed model incorporates DSConv as a lightweight convolutional prior to enhance feature diversity while avoiding excessive parameter growth, thereby achieving a balanced improvement in both performance and computational efficiency.

Ablation experiments are conducted to further provide quantitative and qualitative comparisons between the proposed DSCA mechanism, the linear-attention-based backbone, and the DSConv module. As shown in Table 13, the backbone model with linear attention only (M1) exhibits the largest prediction errors across all datasets, which can be attributed to its limited sensitivity to battery-specific local capacity fluctuations. By enhancing the input representations to the backbone with the DSConv module (M2), the prediction error is significantly reduced, indicating that DSConv effectively captures fine-grained local degradation patterns.

Furthermore, introducing the DSCA mechanism on the backbone while removing DSConv (M3) leads to noticeable performance improvements over both M1 and M2. This result confirms the effectiveness of DSCA in modeling long-term dependencies and capturing global degradation trends. Nevertheless, the absence of DSConv weakens the local learning conditions, thereby limiting the overall performance gains. When DSConv and DSCA are jointly integrated in the proposed model, the highest and most consistent prediction accuracy is achieved across the selected battery datasets.

For example, on the NASA B0005 dataset, the average error is reduced to 0.0058, corresponding to reductions of 50.14%, 19.44%, and 7.94% compared with M1, M2, and M3, respectively. Similar trends are observed on the CALCE dataset: for battery CS2–35, the proposed method reduces the average error from 0.0322 (M1) to 0.0207, achieving a relative reduction of 38.35%. On CS2–36 and CS2–37, the proposed model further decreases the error by 30.34% and 47.04%, respectively, compared to the linear-attention-based backbone.

These results clearly show that the complementary effects of DSConv and DSCA. DSConv reinforces local feature extraction by capturing fine-grained degradation dynamics, while DSCA enriches feature diversity and representation capacity based on linear attention, enabling effective modeling of both long-term degradation trends and short-term capacity fluctuations. Their synergistic integration allows the proposed model to achieve superior accuracy and generalization performance across heterogeneous battery datasets.

Table. 13
Ablation results on NASA and CALCE batteries.

Battery	Model	backbone	DSConv	DSCA	MAE	RMSE	MAPE	Average	Reduction
NASA									
B0005	M1	✓			0.0104	0.0115	0.0130	0.0116	50.14%
	M2	✓	✓		0.0055	0.0089	0.0072	0.0072	19.44%
	M3	✓		✓	0.0049	0.0072	0.0069	0.0063	7.94%
	Proposed	✓	✓	✓	0.0045	0.0067	0.0063	0.0058	–
B0006	M1	✓			0.0251	0.0283	0.0335	0.0289	66.44%
	M2	✓	✓		0.0125	0.0167	0.0156	0.0149	34.90%
	M3	✓		✓	0.0086	0.0146	0.0101	0.0111	12.61%
	Proposed	✓	✓	✓	0.0072	0.0130	0.0090	0.0097	–
B0007	M1	✓			0.0120	0.0152	0.0143	0.0138	34.78%
	M2	✓	✓		0.0095	0.0116	0.0109	0.0107	15.89%
	M3	✓		✓	0.0087	0.0112	0.0098	0.0099	9.09%
	Proposed	✓	✓	✓	0.0076	0.0102	0.0093	0.0090	–
CALCE									
CS2–35	M1	✓			0.0245	0.0306	0.0458	0.0322	38.35%
	M2	✓	✓		0.0202	0.0240	0.0365	0.0269	22.92%
	M3	✓		✓	0.0164	0.0216	0.0289	0.0233	7.03%
	proposed	✓	✓	✓	0.0141	0.0208	0.0273	0.0207	–
CS2–36	M1	✓			0.0253	0.0298	0.0461	0.0314	30.34%
	M2	✓	✓		0.0198	0.0241	0.0401	0.0280	16.07%
	M3	✓		✓	0.0188	0.0236	0.0365	0.0263	10.65%
	proposed	✓	✓	✓	0.0174	0.0212	0.0319	0.0235	–
CS2–37	M1	✓			0.0266	0.0521	0.0465	0.0366	47.04%
	M2	✓	✓		0.0221	0.0395	0.0420	0.0345	36.00%
	M3	✓		✓	0.0198	0.0256	0.0362	0.0272	18.75%
	proposed	✓	✓	✓	0.0155	0.0209	0.0299	0.0221	–

4.4.2. Complexity analysis

In practical applications, it is crucial to evaluate models not only based on accuracy but also by considering computational efficiency and hardware constraints. Key metrics for this evaluation include training cost (FLOPs, training time, and parameter count) and hardware cost (storage size). The comparative results are shown in Table 14.

To ensure a fair comparison, all models are evaluated under identical training and architectural hyperparameters. Specifically, FLOPs, measured using the THOP library's profile function, quantify the floating-point operations required for a single forward pass. Training time, recorded in seconds via Python's time module, reflects the duration needed to train a model on the dataset. The total number of trainable parameters, typically in millions, is determined using the torchsummary library's summary function. Finally, storage size, measured in kilobytes using Python's os.path.getsize function, indicates the space required to store model parameters on a device.

For consistency and comprehensive comparison, all models are trained under two distinct configurations: (1) 1000 epochs with a learning rate of 0.01, and (2) 200 epochs with a learning rate of 0.001. Both settings use a batch size of 128 and a dropout rate of 0.1. To standardize model complexity, Transformer-based architectures employ an embedding dimension of 16, a dense (MLP) dimension of 16, four attention heads, and single-layer encoders and decoders. LSTM-based models use four layers with a hidden dimension of 32. At the same time, all data uses the first 30% of the B0005 battery dataset, with a window size of 5. This dual-setting design allows for a more robust evaluation of model performance under both extensive and limited training conditions.

In CNN-Transformer models, the CNN module consists of a depth-wise separable convolutional layer positioned after the multi-head self-attention module. The convolutional layers are configured with kernel sizes of 3, 2, and 3, and strides of 1, 2, and 1, respectively, followed by ReLU activation. This setup ensures a balanced comparison of computational demands across different architectures.

Under the high-resource setting (1000 epochs), the proposed model achieves a training time of only 11.15 s with 6.45 million parameters, requiring just 0.25 million FLOPs and 30.05 KB of storage. Compared to the CNN-Transformer, which takes 18.48 s and consumes 0.40 million FLOPs with a slightly smaller parameter count (5.54 million), the proposed model achieves a 39.7% reduction in training time and a 37.5% reduction in FLOPs. Against the standard Transformer, the proposed model shows a 31.6% decrease in training time and a 77.3% reduction in FLOPs, while still requiring fewer parameters and less storage. Notably, even compared to LSTM, which has more parameters (8.75 million) and longer training time (8.60 s), the proposed model maintains a more compact architecture and lower computational demand.

Under the low-resource configuration (200 epochs), the efficiency advantages of the proposed model become even more pronounced. It completes training in only 2.60 s, compared to 3.96 s for CNN-Transformer (34.3% faster) and 4.10 s for Transformer (36.6% faster), while maintaining the lowest FLOPs (0.05 M). Furthermore, compared to LSTM, the proposed model reduces FLOPs by 66.7% and cuts training time by over 1.3 s, despite using significantly fewer parameters and storage.

These results clearly show that the proposed model offers a favorable balance between computational efficiency and model capacity, making it especially suitable for deployment in resource-constrained environments where both speed and compactness are critical.

5. Conclusion

This study proposes an innovative deep learning framework for the joint estimation of lithium-ion battery State of Health (SOH) and Remaining Useful Life (RUL). The framework integrates a Deep Spatiotemporal Convolutional Attention (DSCA) mechanism with multi-scale Depthwise Separable Convolutions (DSConv), achieving a balance between computational efficiency and prediction accuracy in feature extraction and time-series modeling. By reformulating the traditional attention computation, the proposed DSCA reduces the computational complexity from $O\left(N^2d\right)$ to $O\left(Nd^2\right)$, significantly lowering the computational cost in large-scale sequence tasks. Meanwhile, the multi-scale DSConv modules effectively capture both long-term degradation trends and short-term capacity recovery behaviors, enhancing the model's capability to represent complex temporal variations during battery aging.

Furthermore, a novel health indicator (HI) selection algorithm based on Pearson and Spearman correlation coefficients (PCC and SCC) is introduced to evaluate and select the most representative indicators, enabling standardized feature selection across different datasets. This approach enhances the model's robustness and generalization in multi-source SOH estimation. To mitigate the effects of short-term capacity regeneration and noise interference, the CEEMDAN decomposition method is employed to extract a smooth and quasi-monotonic capacity degradation trend, providing more stable input data for prediction.

Comprehensive validation on three public datasets demonstrates that the proposed model outperforms existing state-of-the-art methods in terms of prediction accuracy, generalization capability, and computational efficiency. By ensuring high-precision prediction with reduced computational requirements, the proposed framework shows strong potential for real-world deployment in Battery Management Systems (BMS). Specifically, the lightweight model architecture and stable joint SOH–RUL estimation capability enable its implementation in online and embedded BMS environments, where computational resources are limited and prediction robustness is critical. The enhanced degradation trend stability and reduced prediction fluctuations further support its

Table. 14
Efficiency comparison results on B0005.

Models	Training hyperparameters			Model hyperparameters			Performance indicator			
	e	lr	b	d_e	d_d	n	GFLOPs	Training time (s)	Parameters	Storage size(KB)
Proposed	1000	0.01	128	16	16	4	0.25	11.15	6450	30.05
CNN-Transformer	1000	0.01	128	16	16	4	0.40	18.48	5538	39.99
Transformer	1000	0.01	128	16	16	4	1.10	16.29	4625	37.33
LSTM	1000	0.01	128	–	16	4	0.74	8.60	8753	41.12
BMSFormer [2]	1000	0.01	128	16	16	4	0.46	19.83	5330	36.37
Proposed	200	0.001	128	16	16	4	0.05	2.60	6450	30.05
CNN-Transformer	200	0.001	128	16	16	4	0.81	3.96	5538	39.99
Transformer	200	0.001	128	16	16	4	0.20	4.10	4625	37.33
LSTM	200	0.001	128	–	16	4	0.15	1.76	8753	41.12
CNN-Transformer [37]	–	0.00055	–	–	–	–	1.01	806.4	26,100	2235
RB-FA-FF [30]	–	0.0001	32	–	–	–	0.74	–	6,050,000	–
AMN-UKF [72]	–	–	–	–	–	–	2.14	–	231,440	–
MEWOA-VMD-LSTM [73]	–	–	–	–	–	–	5.46	–	269,057	–
SGRU [74]	–	–	–	–	–	–	2.01	–	323,000	–

applicability in real-time health monitoring and decision-making scenarios. In addition, owing to the standardized health indicator selection strategy and robust multi-dataset validation, the proposed framework demonstrates promising adaptability to batteries with different chemical systems and aging characteristics. This suggests that the method can be extended to diverse lithium-ion battery chemistries and application domains without extensive model redesign, facilitating its deployment in practical battery health management and prognostics systems.

Future research will focus on developing lighter and more efficient model architectures while further validating the proposed approach on a wider range of battery materials and chemistries, as well as real-vehicle operational data. Such efforts aim to enhance the model's adaptability and reliability for practical large-scale applications in intelligent energy storage and electric vehicle systems.

CRediT authorship contribution statement

Dexiong Li: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Yan Xing:** Writing – review & editing, Visualization, Project administration, Formal analysis. **Xiaopeng Li:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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